Package 'pigauto'

October 17, 2026

Title Fill in Missing Species Traits Using a Phylogenetic Tree

Version 0.11.0

Description Imputes missing species trait data for comparative analyses by combining three sources of information: phylogenetic similarity (closely related species share similar traits), cross-trait correlations (observed traits inform missing ones), and optional environmental covariates (climate, habitat, geography). Handles continuous measurements, counts, binary variables, ordered categories, unordered categories, bounded proportions, zero-inflated counts, and compositional multi-proportion data in a single call. The default method is a phylogenetic baseline calibrated on held-out data; an optional graph neural network correction, which is needed to use covariates, is blended in through a per-trait gate that retains a baseline fallback. Provides conformal prediction intervals for continuous, count, ordinal, and proportion traits and an experimental analysis-aware multiple-imputation workflow for one missing continuous covariate in Gaussian linear, binomial-logit, and Gaussian random-intercept models, with Rubin pooling limited to fixed effects.
Stochastic graph-network and posterior-tree completions are prediction diagnostics rather than validated inferential imputations.
Bundled datasets include 300-species and 9,993-species bird-trait subsets with matching example phylogenetic trees.
Rubin (1987, ISBN:978-0-471-08705-2); Vovk et al. (2005, ISBN:978-0-387-25061-8); Nakagawa and de Villemereuil (2019) <doi:10.1093/sysbio/syy089>.

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URL https://itchyshin.github.io/pigauto/, https://github.com/itchyshin/pigauto

BugReports https://github.com/itchyshin/pigauto/issues

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`avonet_full` *Full AVONET morphological and ecological trait data for 9,993 bird species*

Description

The full-scale counterpart to `avonet300`: every bird species for which AVONET3 and the bundled example phylogeny agree on both a species label and a complete set of continuous morphometric measurements. The schema is identical to `avonet300` (same trait columns, same factor encodings, same `Species_Key` column) so any code that runs on `avonet300` runs on `avonet_full` with no modification.

Usage

```
avonet_full
```

Format

A data frame with 9,993 rows and 8 variables. Columns match `avonet300`: `Species_Key`, `Mass`, `Beak.Length_Culmen`, `Tarsus.Length`, `Wing.Length`, `Trophic.Level`, `Primary.Lifestyle`, `Migration`.

Details

Unlike the 300-species subset, native AVONET missingness is PRESERVED in the two ecological columns: 4 NAs in `Trophic.Level` and 20 NAs in `Migration`. The four continuous columns are complete by construction.

Use `avonet_full` + `tree_full` to exercise pigauto at real-world scale (the AVONET missingness sweep benchmark uses this dataset). For quick examples or unit tests, prefer the 300-species `avonet300` subset – it is ~30x smaller and loads instantly.

See `avonet300` for the full column schema (same traits and encodings).

Source

Tobias et al. (2022) AVONET: morphological, ecological and geographical data for all birds. *Ecology Letters*, 25, 581-597. Species labels are aligned to the bundled example phylogeny.

See Also

avonet300, tree_full

---

avonet300 *AVONET morphological and ecological trait data for 300 bird species*

Description

A data frame with 300 rows and 8 columns: 4 continuous morphometric traits and 3 ecological traits (2 categorical, 1 ordinal) for demonstrating mixed-type imputation. Species names are in the `Species_Key` column. Set `rownames(df) <- df$Species_Key; df$Species_Key <- NULL` before passing to `preprocess_traits`.

Usage

```
avonet300
```

Format

A data frame with 300 rows and 8 variables:

Species_Key Character. Species name in BirdTree format (spaces replaced by underscores).

Mass Numeric. Body mass (grams).

Beak.Length_Culmen Numeric. Beak length from culmen (mm).

Tarsus.Length Numeric. Tarsus length (mm).

Wing.Length Numeric. Wing length (mm).

Trophic.Level Factor. Dietary category: Carnivore, Herbivore, Omnivore, or Scavenger.

Primary.Lifestyle Factor. Lifestyle category: Aerial, Aquatic, Generalist, Insessorial, or Terrestrial.

Migration Ordered factor. Migration strategy: Resident < Partial < Full.

Source

Tobias et al. (2022) AVONET: morphological, ecological and geographical data for all birds. *Ecology Letters*, 25, 581-597. Species labels are aligned to the bundled example phylogenies.

`build_phylo_graph` *Build a phylogenetic graph representation from a tree*

Description

Computes a symmetric-normalised Gaussian kernel adjacency matrix and Laplacian spectral node features from the cophenetic distance matrix. Results are optionally cached to an `.rds` file.

Usage

```
build_phylo_graph(tree, k_eigen = "auto", sigma_mult = 0.5, cache_path = NULL)
```

Arguments

`tree`	object of class `"phylo"`.
`k_eigen`	integer or `"auto"`. Number of Laplacian eigenvectors to use as node features. When `"auto"` (default), scales with tree size: `min(max(ceiling(n/20), 4), 32)`, then clamped so `k + 1 <= n`. That still gives 4 eigenvectors on trees with 5-80 tips, 8 for 100-160 tips, 15 for 300 tips, and 32 for 640+ tips. On 3- and 4-tip trees the floor of 4 is reduced so the Laplacian eigen-decomposition stays well-defined.
`sigma_mult`	numeric. Bandwidth multiplier (call it s): $\sigma = \text{median}(D) \times s$ (default `0.5`).
`cache_path`	character or `NULL`. Path to an `.rds` cache file. If the file exists and dimensions match, it is loaded instead of recomputing. `NULL` disables caching (default).

Details

The graph is built in three steps: (1) cophenetic distances between all pairs of tips; (2) Gaussian kernel $A_{ij} = \exp(-d_{ij}^2/(2\sigma^2))$ with $\sigma = \text{median}(D) \times s$ (where s is the user-supplied `sigma_mult`); (3) symmetric normalisation $\tilde{A} = D^{-1/2} A D^{-1/2}$ with self-loops added before normalisation.

Spectral node features are the `k_eigen` smallest non-zero eigenvectors of the unnormalised Laplacian, which encode the broad cluster structure of the phylogeny.

Value

A list with:

adj Numeric matrix (n x n). Symmetric-normalised adjacency.

coords Numeric matrix (n x k_eigen). Spectral node features.

n Integer. Number of tips.

sigma Numeric. Bandwidth used.

D Numeric matrix (n x n). Cophenetic (patristic) distances between tips, row/column-ordered by `tree$tip.label`. Returned so downstream functions can reuse it instead of calling `ape::cophenetic.phylo()` again.

D_sq Numeric matrix (n x n). Element-wise squared cophenetic distances (`D * D`). Used by `GraphTransformerBlock` for the learnable per-head Gaussian bandwidth (B2 rate-aware attention). Not freed during `impute()` training — only `D` is freed.

R_phy Numeric matrix (n x n). Phylogenetic correlation matrix `cov2cor(ape::vcv(tree))` (diagonal = 1). Used by `fit_baseline` for BM conditional imputation.

Scaling

For trees with more than 7500 tips, `build_phylo_graph()` uses **RSpectra**'s sparse Lanczos eigensolver (`eigs_sym`) instead of the dense `base::eigen()`. This reduces the spectral step from $O(n^3)$ (hours at $n = 10{,}000$) to $O(n \cdot k \cdot \text{iters})$ (seconds to a minute). The threshold is conservative because the dense eigensolver is competitive on small and mid-size trees, and on pathological ultrametric simulations (`ape::rcoal`) the Laplacian spectrum has tight degenerate clusters that slow down Lanczos until the tree is fairly large. On realistic asymmetric phylogenies with variable branch lengths (e.g. the AVONET BirdTree Stage2 Hackett tree) the sparse path wins much earlier, and at $n = 9{,}993$ it delivers a ~10x speedup over dense. The dense path is kept as the default for smaller trees and as a fallback when **RSpectra** is not installed, or if Lanczos fails to converge. The cophenetic distance matrix is computed once per call and reused for both the adjacency and the spectral features; it is also returned in the result so that downstream consumers (notably `fit_baseline`) can skip recomputation.

Examples

```
set.seed(1)
tree <- ape::rtree(30)
g <- build_phylo_graph(tree, k_eigen = 4)
dim(g$adj)    # 30 x 30
dim(g$coords) # 30 x 4
dim(g$D)      # 30 x 30
```

`calibration_df` *Compute calibration data for probability predictions*

Description

Bins predicted probabilities and computes observed frequencies within each bin. Useful for calibration plots of binary classifiers.

Usage

```
calibration_df(truth, prob, n_bins = 10L)
```

Arguments

`truth`	numeric vector (0/1 or TRUE/FALSE).
`prob`	numeric vector of predicted probabilities in `[0, 1]`.
`n_bins`	integer, number of bins (default 10). Must be a positive integer.

Value

A data.frame with columns `bin_mid`, `obs_freq`, `mean_pred`, `n`.

Examples

```
calibration_df(
  truth = c(0, 0, 1, 1),
  prob = c(0.1, 0.3, 0.7, 0.9),
  n_bins = 2
)
```

check_pigauto *Check whether traits and a tree are ready for pigauto*

Description

Inspect declared inputs without constructing a phylogenetic graph or fitting a model. The check is intended to make input problems and likely compute scale visible before calling `impute()`. Its fingerprint identifies the declared input object for comparison within a workflow; it is not a security hash.

Usage

```
check_pigauto(
  traits,
  tree,
  species_col = NULL,
  trait_types = NULL,
  multi_proportion_groups = NULL,
  covariates = NULL,
  gnn = FALSE
)
```

Arguments

`traits`	A data.frame of traits, with row names or a species column.
`tree`	A `phylo` tree.
`species_col`	Optional column naming species identities.
`trait_types`	Optional named type overrides passed to `preprocess_traits()`.
`multi_proportion_groups`	Optional named composition groups passed to `preprocess_traits()`.
`covariates`	Optional environmental covariates passed to `preprocess_traits()`. Non-finite values are an error.
`gnn`	Whether the planned imputation will use the graph neural network. The default `FALSE` skips the torch runtime probe, matching `impute()`'s default. Set `TRUE` to include the torch runtime in the preflight.

Value

A list of class `pigauto_check` containing a schema version, status, stable input fingerprint, input summaries, and structured messages.

Examples

```
tree <- ape::read.tree(text = "((a:1,b:1):1,c:2);")
traits <- data.frame(x = c(1, NA, 3), row.names = tree$tip.label)
check_pigauto(traits, tree)
```

compare_methods *Compare BM baseline and pigauto methods across replicates*

Description

Runs a full comparison of the BM baseline and pigauto (with attention + calibration) over multiple random seeds. Returns a tidy data.frame suitable for plotting or downstream analysis.

Usage

```
compare_methods(
  data,
  tree,
  splits = NULL,
  seeds = NULL,
  epochs = 500L,
  verbose = TRUE,
  gnn = TRUE,
  ...
)
```

Arguments

data	pigauto_data object.
tree	phylo object.
splits	pre-computed splits (applied to all reps) or NULL to create fresh splits per seed.
seeds	optional integer vector of random seeds for replication. The default NULL performs one run using the current RNG stream.
epochs	number of training epochs.
verbose	logical.
gnn	logical. Whether the pigauto arm trains the GNN. Default TRUE, so this function keeps comparing the baseline with the GNN-corrected model even though fit_pigauto and impute default to gnn = FALSE.
...	additional arguments passed to fit_pigauto.

Details

For each seed the function:

1. Creates train/val/test splits.
2. Fits the phylogenetic BM baseline.
3. Fits pigauto (with attention and calibration enabled by default).
4. Evaluates both methods on the test split.
5. Collects results into a single data.frame.

The default comparison trains the optional torch GNN. The example checks that torch can create a tensor before fitting; it reports a skip when the Lantern runtime is unavailable.

Value

A data.frame with columns: method, trait, type, metric, value, rep.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
torch_ready <- requireNamespace("torch", quietly = TRUE) &&
  isTRUE(tryCatch(torch::torch_is_installed(), error = function(e) FALSE)) &&
  isTRUE(tryCatch({
    probe <- torch::torch_tensor(1)
    as.numeric(probe$item()) == 1
  }, error = function(e) FALSE))
if (torch_ready) {
  cmp <- compare_methods(preprocess_traits(traits, tree), tree, seeds = 1L,
                         epochs = 5L, verbose = FALSE, gnn = TRUE)
  aggregate(value ~ method + trait + metric, data = cmp, FUN = mean)
} else {
  message("Skipping GNN comparison: install torch and its Lantern runtime.")
}
```

---

completed_data *Extract completed trait data*

Description

Extract completed trait data

Usage

```
completed_data(result)
```

Arguments

result	A pigauto_result.

Value

Exactly the stored result$completed data.frame.

Examples

```
## Not run: completed_data(result)
```

confusion_matrix *Compute a confusion matrix for categorical or binary predictions*

Description

Compute a confusion matrix for categorical or binary predictions

Usage

```
confusion_matrix(truth, predicted, levels = NULL)
```

Arguments

truth	factor or character vector of true classes.
predicted	factor or character vector of predicted classes.
levels	character vector of class levels (default: union of truth and predicted levels).

Value

A list with:

table Confusion matrix (rows = truth, columns = predicted).

accuracy Overall accuracy.

per_class Data.frame with per-class precision, recall, F1.

Examples

```
confusion_matrix(
  truth = c("a", "a", "b", "b"),
  predicted = c("a", "b", "b", "b")
)
```

cross_validate *k-fold cross-validation for pigauto trait imputation*

Description

Performs stratified k-fold cross-validation by rotating which cells serve as the test set. Returns per-fold and aggregated metrics.

Usage

```
cross_validate(
  data,
  tree,
  k = 5L,
  seeds = NULL,
  epochs = 500L,
  verbose = TRUE,
  ...
)
```

Arguments

data	pigauto_data object (output of preprocess_traits).
tree	phylo object.
k	integer. Number of folds (default 5).
seeds	optional integer vector. Seeds for replicate runs. The default NULL performs one run using the current RNG stream.
epochs	integer. Training epochs per fold (default 500).
verbose	logical. Print progress (default TRUE).
...	Additional arguments passed to fit_pigauto (e.g. hidden_dim, k_eigen, use_attention). cross_validate() precomputes one graph shared across folds and uses k_eigen = 8L for that graph when k_eigen is omitted. An explicit k_eigen is used for the shared graph and forwarded to fit_pigauto().

Value

A list of class "pigauto_cv" with:

results Data.frame with columns: fold, rep, trait, type, metric, value.

summary Data.frame with mean and sd across folds/reps for each trait + metric.

conformal_coverage Data.frame of coverage per trait across folds (if available).

k Number of folds.

n_reps Number of replicates.

Examples

```
data(avonet300, tree300, package = "pigauto")
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(12L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
cv <- cross_validate(preprocess_traits(traits, tree), tree, k = 3L,
                     seeds = 1L, epochs = 1L, verbose = FALSE,
                     k_eigen = 4L, hidden_dim = 8L,
                     use_transformer_blocks = FALSE)
print(cv)
summary(cv)
```

---

ctmax_sim *Simulated multi-observation-per-species CTmax data*

Description

A simulated dataset mimicking a thermal tolerance study where each species has multiple measurements of critical thermal maximum (CTmax) taken at different acclimation temperatures. The data-generating process is

Usage

```
ctmax_sim
```

Format

A data frame with 1,464 rows and 3 variables:

species Character. Species name matching `tree300` tips.

acclim_temp Numeric. Acclimation temperature (degrees C). This is an observation-level covariate that varies within species.

CTmax Numeric. Critical thermal maximum (degrees C). Contains NA values for unobserved species and MCAR missingness.

Details

$$CTmax_{ij} = 38 + phylo_i + 0.50 \times acclim_temp_{ij} + \epsilon_{ij}$$

where $phylo_i$ follows Brownian motion on `tree300`, the within-species acclimation response ratio is 0.50, and $\epsilon_{ij} \sim N(0, 1.5)$. Thirty percent of species are entirely unobserved (all CTmax values are NA), and an additional 15\ observations are missing at random.

This dataset demonstrates pigauto's multi-observation and observation-level covariate support. Use with `tree300` and set `species_col = "species"` in `impute`.

Source

Simulated. See `data-raw/make_ctmax_sim.R` for the generation script.

See Also

`tree300`, `impute`

`evaluate` *Evaluate a fitted pigauto model on its test set*

Description

Computes type-specific performance metrics for each trait. By default the model is evaluated on the test split stored in the fit object, but an alternative `data` / `splits` can be supplied.

Usage

```
evaluate(fit, data = NULL, splits = NULL)
```

Arguments

`fit`	pigauto_fit object.
`data`	pigauto_data object (default: `NULL`, uses the training data stored in the fit via `predict()`).
`splits`	splits object (default: `NULL`, uses `fit$splits`).

Details

Metrics by trait type:

continuous RMSE, Pearson r, MAE

count RMSE, MAE, Pearson r

binary Accuracy, Brier score

categorical Accuracy (overall)

ordinal RMSE (on integer scale), Spearman rho

When conformal scores are present in the fit, conformal coverage at the 95\ ordinal traits.

When the fit includes a baseline, baseline metrics are appended with `method = "baseline"` for direct comparison.

Value

A `data.frame` with columns: `method`, `trait`, `type`, `metric`, `value`, `n_test`.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
pd <- preprocess_traits(traits, tree)
splits <- make_missing_splits(pd$X_scaled, trait_map = pd$trait_map)
fit <- fit_pigauto(pd, tree, splits = splits, epochs = 5L,
                   verbose = FALSE)
eval_df <- evaluate(fit)
eval_df[eval_df$metric == "rmse", ]
```

---

evaluate_imputation *Evaluate imputation performance against known values*

Description

Computes type-specific metrics for each trait on the validation and test splits. When a `trait_map` is supplied, metrics are dispatched per trait type; otherwise the function falls back to continuous-only metrics (RMSE, Pearson r, 95% coverage).

Usage

```
evaluate_imputation(pred, truth, splits, pred_se = NULL, trait_map = NULL)
```

Arguments

pred	predicted values: either a numeric matrix in latent scale (same dimensions as truth), or a "pigauto_pred" object from predict.pigauto_fit.
truth	numeric matrix of true values in latent scale (from pigauto_data$X_scaled).
splits	list (output of make_missing_splits).
pred_se	numeric matrix of prediction SEs (same scale as pred). Used for 95\ when pred is a pigauto_pred (uses pred$se).
trait_map	list of trait descriptors (from pigauto_data). If NULL and pred is not a pigauto_pred, the v0.1 all-continuous evaluation is used.

Details

Metrics by trait type:

continuous RMSE, Pearson r, 95\ supplied)

proportion RMSE, Pearson r, 95\ supplied)

count RMSE, MAE, Pearson r

ordinal RMSE, Spearman rho

binary Accuracy, Brier score

categorical Accuracy

zi_count RMSE, MAE, Pearson r, zero-accuracy, Brier score (on gate)

For binary and categorical traits the function accepts either a pigauto_pred object (preferred, gives access to probabilities) or raw matrices (latent scale).

Value

A data.frame with columns split, trait, type, n, and type-specific metric columns.

Examples

```
truth <- matrix(c(1, 2, 3, 4), nrow = 2L)
splits <- list(test_idx = 1L, val_idx = 2L, n = 2L, p = 2L)
evaluate_imputation(truth, truth, splits)
```

---

fit_baseline *Fit the phylogenetic baseline*

Description

Dispatches to pigauto's phylogenetic baseline machinery and returns imputed latent-scale means and standard errors for every species.

Usage

```
fit_baseline(
  data,
  tree,
  splits = NULL,
  model = "BM",
  graph = NULL,
  multi_obs_aggregation = c("hard", "soft"),
  lambda_mode = c("estimate", "fixed_1", "cv", "bayes"),
  lambda_fixed = NULL,
  discrete_lambda = c("estimate", "fixed_1"),
  em_iterations = 0L,
  em_tol = 0.001,
  em_offdiag = FALSE,
  joint_solver = c("inhouse", "rphylopars"),
  predict_method = c("auto", "exact", "per_column"),
  joint_refine_iter = 0L,
  predict_route = NULL,
  seed = NULL
)
```

Arguments

data	object of class "pigauto_data".
tree	object of class "phylo".
splits	list (output of make_missing_splits) or NULL.
model	character. Evolutionary model: "BM" (default) or "OU".
graph	optional list returned by build_phylo_graph. When supplied, graph$D (cophenetic distances) is reused for label propagation and graph$R_phy (phylogenetic correlation matrix) is reused for BM imputation, avoiding duplicate $O(n^2)$ allocations. When NULL (default), both matrices are computed here.
multi_obs_aggregation	character. How to aggregate multiple observations per species before the Level-C joint baseline: "hard" (default) thresholds binary proportions at 0.5 and uses argmax for categorical, matching Phase 10 behaviour. "soft" preserves species-level proportions and dispatches the truncated-Gaussian soft E-step (estep_liability_bina so that intermediate class frequencies contribute fractional liability evidence. Only relevant for multi-obs data with binary or categorical traits when the Level-C joint baseline is active.
lambda_mode	character. Pagel-lambda mode for the CONTINUOUS-FAMILY baseline (continuous, count, proportion, zi_count magnitude columns; NOT ordinal, which the threshold-joint path treats as a liability column via estep_liability_ordinal() – see "Per-type lambda dispatch" in Details). "estimate" (default, S4) fits each continuous-family column's own Pagel's lambda via profile REML; when the joint MVN or threshold-joint baseline fires, this now runs INSIDE that joint fit (R/joint_mvn_solver.R's per-column lambda_cols) rather than being discarded in favour of a separate per-column re-fit. "fixed_1" preserves the classic Brownian correlation matrix (lambda = 1 everywhere). "cv" and "bayes" have no joint analogue and force the per-column BM path for continuous-family columns (as before); binary/ordinal/categorical/ zi_gate columns are unaffected

by lambda_mode in every case; their lambda is governed by discrete_lambda (below), and is 1 when discrete_lambda = "fixed_1". **Covariate caveat**: when data$covariates is supplied, the per-column path switches to bm_impute_col_with_cov(), which accepts a numeric lambda or "estimate" but not "cv" / "bayes"; those two modes are silently ignored (fit at lambda = 1) for BM-eligible columns and a warning is emitted.

lambda_fixed — optional named numeric vector (names = latent column names, i.e. colnames(data$X_scaled)) giving a FIXED lambda per continuous-family column, overriding lambda_mode entirely for those columns (spec 4.5: reproduce a previous fit's exact per-trait lambda at predict time without re-estimating – typically supply that fit's own $lambda_per_trait). Columns not present in lambda_fixed keep their lambda = 1 default. NULL (default) means every continuous-family column follows lambda_mode normally.

discrete_lambda — character, "estimate" (default) or "fixed_1". With "estimate", the binary, ordinal and zero-inflated-count gate liability columns of the threshold-joint baseline, and the one-vs-rest column of each categorical class, get their own Pagel's lambda, so a weak-signal discrete trait shrinks toward its prevalence instead of copying close relatives. "fixed_1" restores the previous behaviour (binary and ordinal at lambda = 1, or at the shared joint lambda on the "exact" route; categorical at 1). Evidence (simulated Brownian-motion and OU data, n = 100 to 1000, 30\ missing; docs/dev-log/discrete-lambda/README.md): estimating discrete lambda alone added about 0.01 to 0.04 discrete accuracy at Pagel's lambda of 0.3 or below; together with safety_floor = FALSE and phylo_signal_gate = FALSE (also the defaults) the gain was 0.04 to 0.12, and binary and categorical accuracy matched or beat BACE's. Costs: with lambda near 1 and n >= 1000, accuracy -0.002 to -0.005 and Brier score +0.010 to +0.016; on AVONET300's categorical traits, Brier +0.014 to +0.020 with accuracy unchanged or better. Ordinal accuracy still trails BACE's at lambda = 0.3. The categorical one-vs-rest lambdas are not reported in $lambda_per_trait. Recorded in the returned list and in a fit's model_config; binary and ordinal lambdas are replayed by lambda_fixed.

em_iterations — integer. Number of Phase 6 EM iterations for the threshold-joint baseline (binary + ordinal + OVR categorical). Default 0L disables the EM loop and preserves v0.9.1 output byte-for-byte. When >= 1, the BM rate Σ learned by the in-house joint solver (R/joint_mvn_solver.R) at iteration k is fed back as the per-trait prior SD at iteration $k+1$, up to em_iterations times or until em_tol convergence. em_iterations = 1L is a degenerate single-pass run and produces the same baseline output as 0L; >= 2L is needed for actual iteration. Only affects the threshold-joint path (continuous-only traits pass through the existing joint MVN path unchanged).

em_tol — numeric. Relative-Frobenius convergence tolerance for the Phase 6 / 7 EM loop. Early-stops when $||\Sigma_k - \Sigma_{k-1}||_F / ||\Sigma_{k-1}||_F <$ em_tol. Default 1e-3.

em_offdiag — logical. Phase 7 opt-in: when TRUE AND em_iterations >= 2L, each liability cell's prior at iteration $k+1$ is the conditional-MVN (μ, sd) given the posterior liability of other traits at iteration k, using the full off- diagonal entries of Σ. Binary + ordinal only (OVR categorical stays on Phase 6 diagonal). Default FALSE preserves Phase 6 behaviour.

joint_solver — character. Which solver estimates the joint Sigma / posterior for the joint MVN, threshold-joint, and OVR categorical baselines. "inhouse" (default) uses the single-pass in-house solver (R/joint_mvn_solver.R); under lambda_mode =

"fixed_1" it is byte-identical to prior releases, but lambda_mode = "estimate" (the current default) is new behaviour, not a byte-compatibility guarantee. "rphylopars" delegates to Rphylopars::phylopars(); under "estimate" this calls it with model = "lambda" (substantially slower than model = "BM"), with automatic fallback to "inhouse" on failure or implausible output (a plausibility guard that also fires under lambda_mode = "fixed_1" – see NEWS). Only affects the joint MVN / threshold-joint / OVR categorical paths above; ignored when those paths don't fire. lambda_mode does NOT disable the continuous-only joint MVN path: both fit_joint_mvn_baseline() and fit_joint_threshold_baseline() accept a lambda_mode / lambda_fixed argument and estimate lambda inside the joint fit via lambda_cols – see "Per-type lambda dispatch" in Details. Only "cv" / "bayes" force continuous-family columns off the joint path entirely (no joint analogue for those two modes).

predict_method character. Prediction route for the in-house joint solver. "auto" (default, S5b/S5c) fits the baseline once with the "exact" route and once with the "per_column" route, then for each TRAIT (each trait_map entry; a categorical trait's K latent columns and a zi_count trait's gate + magnitude columns are chosen together) picks whichever route has the lower loss on that trait's cells: mean squared error on the z-scored latent scale for continuous/count/proportion/ordinal/zi_count magnitude, mean log-loss of plogis(mu) against the observed 0/1 truth for binary/zi_count gate, and mean multinomial log-loss for categorical. **Which cells the choice sees**: a trait's held-out validation rows (species in multi-observation data) are split into two halves (seeded via seed, or the ambient RNG state when seed = NULL) when the trait has at least 38 of them and the two candidate fits predict it differently. The route choice sees only half A; half B is reserved so fit_pigauto()/impute() calibrate the GNN gate and compute conformal scores on rows that never informed the choice. Below 38 rows each half would keep fewer than the 19 a 95\ split-conformal interval needs, so the same rows both choose the route and calibrate; when both fits agree, no choice is made and all rows calibrate. With fewer than 5 validation cells, or on a genuine tie, a trait keeps "exact", as it does when splits is NULL (no validation cells at all makes every trait "exact"). The chosen route per trait is returned as $predict_method_by_trait. "exact" uses the full cross-trait conditional mean and variance of vec(L) ~ MVN(0, Sigma %x% R(lambda_block)) in the sparse precision form (Hadfield & Nakagawa, 2010), with each column centred at its own GLS phylogenetic mean at lambda_block before the solve (mean-model consistency; see docs/dev-log/exact-default/). Discrete liability columns (binary, zi gate, ordinal-via-OVR synthetic columns) share lambda_block under "exact" rather than staying fixed at lambda = 1, so the whole joint fit uses one internally-consistent R(lambda_block). Falls back to "per_column" above roughly 20000 unknown cells (roughly 4000 species at 5 traits), on a singular/unusable Sigma, or when fewer than 2 joint columns or no Henderson sparse precision are available; the fallback prints a one-time message() (not a warning) per R session when predict_method was left at its default, or a warning() every time when "exact" was requested explicitly. "per_column" retains the original per-column conditional prediction route (each column's own posterior, no cross-trait borrowing in the prediction step; se is that column's own conditional SE). Neither option changes covariance estimation or the "rphylopars" solver.

joint_refine_iter
integer, default 0L. Enables cross-trait refinement of the joint baseline's cell imputations using the estimated Sigma (the in-house solver's max_iter EM cell-

refinement; R/joint_mvn_solver.R). 0L preserves current behaviour byte-for-byte. The refinement is guarded: the Sigma step must shrink each iteration, or the loop rolls back to the last good iterate and sets $diverged. **Has no effect on any trait predicted by the "exact" route**: the exact conditional solve returns its analytic posterior directly and never enters this EM loop. Under the default predict_method = "auto", joint_refine_iter therefore only ever applies to traits that "auto" routes to "per_column" (see $predict_method_by_trait below).

predict_route — optional named character vector (names = trait names, i.e. vapply(data$trait_map, function(tm) tm$name, character(1)); values "exact" or "per_column"). Forces the prediction route for each named trait, bypassing the "auto" validation-loss comparison entirely (and overriding predict_method, if a value other than "auto" was also supplied). Traits not named default to "exact". A name that matches no trait in data$trait_map is ignored, with a warning (once per call). Used internally to replay a validation-split "auto" choice onto a production refit that has no validation cells (e.g. fit_pigauto()'s baseline_full, fit with splits = NULL) so that refit reuses the SAME per-trait route rather than re-deciding with zero validation evidence. NULL (default) disables forcing; predict_method governs normally.

seed — optional integer. Seeds the deterministic half-A (route choice) / half-B (reserved for fit_pigauto()/impute()'s gate calibration and conformal scoring) split of each trait's validation cells under predict_method = "auto" with real validation cells (see ""auto"" above). NULL (default) leaves the split to whatever the ambient RNG state is at call time, matching this package's other optional seed arguments. Ignored when predict_method is "exact" / "per_column", when predict_route is supplied, or when there are no validation cells.

Details

When splits is supplied the val and test cells are masked to NA before fitting, so the baseline is evaluated under the same conditions as fit_pigauto.

Continuous-family columns use Brownian-motion conditional MVN baselines on the phylogenetic correlation matrix, either independently or through the joint MVN path when the data and optional dependencies support it. Binary, ordinal, categorical, and zero-inflated gate columns use the appropriate label-propagation or threshold/liability baseline candidates, with per-column fallbacks when a joint path is not available.

Covariates and the joint baseline (P1-8): data$covariates is only used by the per-column BM path (bm_impute_col_with_cov()). The joint MVN and threshold-joint (Rphylopars) baselines do not accept a covariate design matrix, so when a joint path is selected (BM-eligible columns >= 2, or binary/ordinal cols present, with Rphylopars available) any supplied covariates are ignored for the BASELINE and a warning is emitted; covariates still reach the GNN correction via fit_pigauto regardless of which baseline path fires.

Per-type lambda dispatch (arc/lambda-per-type; corrected in S4/S6, feat/joint-lambda-default): lambda_mode only ever governs the baseline for CONTINUOUS-FAMILY columns (continuous, count, proportion, zi_count magnitude) – NOT ordinal. Binary, ordinal, categorical, and zero-inflated gate columns always stay at lambda = 1 in EVERY path (threshold-joint, OVR-categorical, label propagation, and the per-trait ordinal path-selection fallback below), regardless of lambda_mode – there is no discrete-trait analogue of Pagel's lambda, and previously forcing these columns onto label propagation any time lambda_mode != "fixed_1" cost 19pp of Trophic.Level accuracy on AVONET (0.789 -> 0.600; see docs/dev-log/2026-08-16-external-comparison-results.md).

When lambda_mode != "fixed_1" and the threshold-joint baseline fires for a dataset with binary/ordinal AND continuous-family columns, the joint liability fit's own continuous-column OUTPUT is now USED (each continuous-family column gets its own lambda_k via the joint solver's lambda_cols), rather than being discarded for a separate lambda-aware per-column BM re-fit as in the pre-S4 design. Ordinal liability columns inside that same joint fit, and the BM-via-MVN / K-class-OVR-LP alternatives the "Per-trait ordinal path selection" block below compares against it, all stay at lambda = 1.

Value

A list with:

mu Numeric matrix (n_species x p_latent), baseline means in latent scale.

se Numeric matrix (n_species x p_latent), standard errors.

path Named character vector, one entry per trait_map entry (name = trait/group name), recording which dispatch branch produced that trait's baseline: "joint_mvn" (Phase 2 continuous-only joint), "threshold_joint" (Phase 3/5 binary + ordinal + continuous-family liability joint), "ovr_categorical" (Phase 6 one-vs-rest K-fits), "per_column_bm" (per-column Brownian-motion conditional MVN, including the lambda-aware path and the per-trait ordinal fallback that beats threshold-joint on held-out val MSE), "multi_proportion_bm" (per-component per_column_bm on the CLR columns of a multi_proportion group – same kernel as "per_column_bm", labelled separately because multi_proportion is architecturally distinct), "label_propagation" (phylogenetic label propagation for binary/categorical/ordinal, and the per-trait ordinal fallback that beats the other two candidates), "zi_gate_lp" (label propagation for a zero-inflated count's gate column), or "zi_mag_constant" (global mean/sd fallback for a zero-inflated count's magnitude column when fewer than 5 non-zero observations exist). For zi_count traits (2 latent columns: gate then magnitude), path reports the GATE column's dispatch – the magnitude column can independently land on "joint_mvn", "threshold_joint", "per_column_bm", or "zi_mag_constant" and is not separately surfaced here.

lambda_per_trait Named numeric vector (length p_latent, names = colnames(data$X_scaled)), the lambda actually used for each latent column: 1 for every column not eligible for lambda estimation (all discrete columns; continuous-family columns under lambda_mode %in% c("cv", "bayes")), else the estimated or fixed value.

lambda_block Numeric scalar, the shared lambda used internally by whichever joint fit ran for its Sigma M-step / opt-in exact conditional / opt-in EM refine (see R/joint_mvn_solver.R's $lambda_block); NA when no joint fit ran.

lambda_mode Character, echoes the resolved lambda_mode argument.

predict_method_used Character scalar, "exact", "per_column", or "auto". Under a concrete predict_method ("exact" / "per_column"), aggregated across every joint fit that ran: "exact" only if every joint fit achieved exact; "per_column" if any fell back, if none ran (a pure per-column baseline), or if every joint fit used joint_solver = "rphylopars" (no exact/per_column dichotomy there). Under predict_method = "auto" (or a forced predict_route), always "auto"; see $predict_method_by_trait for the resolved per-trait routes.

predict_method_by_trait Named character vector (names = trait names), one entry per trait_map entry, giving the route ("exact" or "per_column") actually used for that trait's cells. Under a concrete predict_method, every entry equals predict_method_used. Under "auto" (or a forced predict_route), the per-trait validation choice (see predict_method's "auto" entry above).

Examples

```
data(avonet300, tree300, package = "pigauto")
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
pd     <- preprocess_traits(traits, tree)
splits <- make_missing_splits(pd$X_scaled, trait_map = pd$trait_map)
bl     <- fit_baseline(pd, tree, splits)
```

`fit_pigauto` *Fit a pigauto model for trait imputation*

Description

Trains a pigauto model: a gated ensemble of a phylogenetic baseline and an attention-based graph neural network correction, implemented as an internal torch module (`ResidualPhyloDAE`; "Residual" here refers to the ResNet-style skip connections in the GNN layers, not to a statistical residual). For continuous, count, and ordinal traits the baseline is Brownian motion (phylogenetic correlation matrix); for binary and categorical traits it is phylogenetic label propagation. Supports all eight trait types via a unified latent space.

Usage

```
fit_pigauto(
  data,
  tree,
  splits = NULL,
  graph = NULL,
  baseline = NULL,
  gnn = FALSE,
  baseline_full = NULL,
  hidden_dim = 64L,
  k_eigen = "auto",
  n_gnn_layers = 2L,
  gate_cap = 0.8,
  use_attention = TRUE,
  use_transformer_blocks = TRUE,
  n_heads = 4L,
  ffn_mult = 4L,
  use_trait_attention = FALSE,
  n_trait_heads = 2L,
  trait_embed_dim = 32L,
  dropout = 0.1,
  lr = 0.003,
  weight_decay = 1e-04,
  epochs = 3000L,
  corruption_rate = 0.55,
  corruption_start = 0.2,
  corruption_ramp = 500L,
```

```
  refine_steps = 8L,
  lambda_shrink = 0.03,
  lambda_gate = 0.01,
  warmup_epochs = 200L,
  edge_dropout = 0.1,
  eval_every = 100L,
  patience = 10L,
  clip_norm = 1,
  conformal_method = c("split", "bootstrap", "mondrian"),
  conformal_bootstrap_B = 500L,
  conformal_split_val = FALSE,
  gate_method = c("cv_folds", "median_splits", "single_split"),
  gate_splits_B = 31L,
  gate_cv_folds = 5L,
  safety_floor = FALSE,
  phylo_signal_gate = FALSE,
  phylo_signal_threshold = 0.2,
  phylo_signal_method = c("lambda", "blomberg_k"),
  min_val_cells = 20L,
  lambda_mode = c("estimate", "fixed_1", "cv", "bayes"),
  discrete_lambda = c("estimate", "fixed_1"),
  joint_solver = c("inhouse", "rphylopars"),
  predict_method = c("auto", "exact", "per_column"),
  joint_refine_iter = 0L,
  verbose = TRUE,
  seed = NULL
)
```

Arguments

data	object of class "pigauto_data".
tree	object of class "phylo".
splits	list (output of make_missing_splits) or NULL.
graph	list (output of build_phylo_graph) or NULL.
baseline	list (output of fit_baseline) or NULL.
gnn	logical. When TRUE, trains the attention-based GNN correction as described above. When FALSE (default), no GNN is constructed or trained and fit_pigauto() makes **zero** torch:: calls – the fit is the phylogenetic baseline alone, optionally re-weighted against a grand-mean floor. safety_floor and phylo_signal_gate keep their usual semantics: the blend collapses to $r_{BM} \cdot \mu_{BM} + r_{MEAN} \cdot \mu_{MEAN}$ (the GNN corner degenerates to the baseline, so its calibrated weight is folded into r_{BM}). With the defaults (safety_floor = FALSE, phylo_signal_gate = FALSE) this is the pure traditional-stats arm. conformal_method = "mondrian" is not supported when gnn = FALSE (its locality statistic conditions on a calibrated GNN prediction surface that does not exist here) and raises an error. User covariates enter pigauto only through the GNN, so under gnn = FALSE they are ignored (with a warning). Without a validation split (splits = NULL) the fit is pure baseline ($r_{BM} = 1$) and carries no conformal scores. Default FALSE since pigauto 0.11.0.9001: on simulated and real data at the current baseline defaults the GNN did not lower imputation error and took 20 to 90 times longer to fit (docs/dev-log/arc/2026-10-02-campaign-gnn-rerun.md). Set gnn =

TRUE to train it, for example to use covariates, which only the GNN uses, or to restore the behaviour of earlier versions.

baseline_full — list (output of fit_baseline with splits = NULL) or NULL. Only used when gnn = FALSE: the production-mode baseline, fit on ALL observed cells (no val/test hold-out), used by predict.pigauto_fit() for ordinary (non-evaluation) predictions. baseline itself always stays the held-out fit and is what every scorer (val_rmse, test_rmse, conformal_scores, evaluate()) reads. When NULL (default) and gnn = FALSE, it is computed internally via fit_baseline(data, tree, splits = NULL, ...); ignored when gnn = TRUE unless explicitly supplied (e.g. by impute()), in which case it is stored but not used for training.

hidden_dim — integer. Hidden layer width (default 64).

k_eigen — integer or "auto". Number of spectral node features. The default "auto" scales with tree size; an integer specifies the dimension directly.

n_gnn_layers — integer. Number of graph message-passing layers (default 2). Each layer has its own learnable alpha gate, layer normalisation, and ResNet-style skip connection.

gate_cap — numeric. Upper bound for the per-column blend gate (default 0.8). Safety comes from regularisation, not the cap.

use_attention — logical. Use attention in the GNN layers (default TRUE).

use_transformer_blocks — logical. Replace the legacy attention stack with pre-norm transformer-encoder blocks (multi-head attention

- FFN + two residual skips). Default TRUE. Set FALSE to reconstruct pre-v0.9.0 fits (single-head attention with a learnable alpha gate per layer).

n_heads — integer. Number of attention heads when use_transformer_blocks = TRUE (default 4). Each head learns its own phylogenetic bandwidth (B2 rate-aware attention).

ffn_mult — integer. Feed-forward width multiplier inside each transformer block (default 4, giving hidden_dim * 4).

use_trait_attention — logical. Opt-in within-row cross-trait self-attention (B3, v0.9.3). When TRUE (default FALSE), the model builds per-trait tokens from each row's latent values (linear projection + learnable positional embedding), applies one multi-head self-attention block over the trait sequence, mean-pools to a trait_embed_dim feature, and concatenates it alongside (x, coords, covs) at the encoder input. Intended for trait sets with within-row functional coupling not represented by the selected baseline. Assess this opt-in architecture with held-out evaluation on the intended data. Default FALSE preserves v0.9.2 behaviour exactly.

n_trait_heads — integer. Number of attention heads in the within-row self-attention block when use_trait_attention = TRUE. Default 2. Ignored when use_trait_attention = FALSE.

trait_embed_dim — integer. Embedding dim per trait token in the within-row self-attention block. Default 32. Ignored when use_trait_attention = FALSE.

dropout — numeric. Dropout rate (default 0.10).

lr — numeric. AdamW learning rate (default 0.003).

weight_decay — numeric. AdamW weight decay (default 1e-4).

epochs — integer. Maximum training epochs (default 3000).

corruption_rate
: numeric. Final corruption fraction if corruption_ramp > 0; otherwise the fixed corruption rate per epoch (default 0.55).

corruption_start
: numeric. Initial corruption fraction for the curriculum schedule (default 0.20). Ignored if corruption_ramp = 0.

corruption_ramp
: integer. Epochs over which corruption linearly ramps from corruption_start to corruption_rate (default 500). Set to 0 for fixed corruption.

refine_steps
: integer. Iterative refinement steps at inference (default 8). Gate calibration and conformal scoring use the same number of steps so the calibrated surface matches predict.

lambda_shrink
: numeric. Weight on the shrinkage penalty ||delta - baseline||^2 that keeps the GNN correction close to the phylogenetic baseline (default 0.03).

lambda_gate
: numeric. Weight on the gate regularisation penalty that pushes learnable gates toward zero. Prevents gates from staying open when the GNN provides no useful correction (default 0.01).

warmup_epochs
: integer. Linear learning-rate warmup over the first N epochs (default 200). After warmup, a cosine schedule decays the LR to 1e-5.

edge_dropout
: numeric. Fraction of adjacency edges randomly zeroed each training epoch for graph regularisation (default 0.1). Set to 0 to disable.

eval_every
: integer. Evaluate on val every N epochs (default 100).

patience
: integer. Early-stopping patience in eval cycles (default 10).

clip_norm
: numeric. Gradient clip norm (default 1.0).

conformal_method
: character. How the conformal residual score is estimated from held-out validation cells. "split" (default, backward-compatible) takes a sample quantile; "bootstrap" averages quantiles from conformal_bootstrap_B bootstrap resamples; and "mondrian" stratifies validation cells by phylogenetic sampling locality before estimating a score per stratum. These choices support nominal held-out diagnostics, not package-certified coverage. "mondrian" is single-observation only because its locality is species-level. If either stratum is too small, that trait uses the global "split" score instead.

conformal_bootstrap_B
: integer. Bootstrap resamples used when conformal_method = "bootstrap"; default 500. Ignored otherwise.

conformal_split_val
: logical. Default FALSE: gate selection and residual-score estimation share validation cells. When TRUE, each eligible latent column separates those tasks into calibration and residual-estimation subsets. This separation does not certify package-wide coverage. Columns without enough validation cells retain the shared-data path. With either setting, use cross_validate() and report held-out diagnostics for the intended data.

gate_method
: character. How the per-trait calibrated gate is chosen. "single_split" runs the grid search on a single random half-A / half-B split of the val rows; "median_splits" repeats the whole procedure for gate_splits_B random splits and takes the median best_g. "cv_folds" (default since PR #102, 2026-05-17) partitions val cells into gate_cv_folds (default 5) deterministic non-overlapping folds and runs the grid + half-B-verify procedure once per fold (training set = K-1 folds,

held-out = remaining fold), taking the componentwise median of K winning weight vectors. "cv_folds" uses larger training sets per split (K-1/K vs 1/2 in median_splits) and has a standard cross-validation interpretation, motivated by the open val→test drift observed on 4/32 binary cells in the discrete-bench memo. "median_splits" repeats the split procedure before taking a componentwise median. Assess calibration settings on held-out data.

gate_splits_B
: integer. Random splits used when gate_method = "median_splits"; default 31 (odd so the median is well-defined).

gate_cv_folds
: integer. Number of CV folds when gate_method = "cv_folds"; default 5, must be >= 2. Capped at n_val per trait so each fold has at least 1 cell. When effective K < 2 (e.g. n_val = 1), the code falls back to a single split.

safety_floor
: logical. Default FALSE (it was TRUE from v0.9.1.9002 until 0.11.0.9002). When TRUE, post-training calibration searches a 3-way simplex of BM, GNN, and grand-mean candidates. Because the grand-mean corner is always in the grid, the selected candidate cannot be worse than that corner on the validation cells under the calibration metric. When FALSE, the v0.9.1 1-D calibration is used exactly (r_MEAN = 0). The default changed because, with Pagel's λ now estimated for every trait type (see discrete_lambda), the floor no longer improved accuracy in any tested setting; at λ of 0 to 0.1 the old default was worse than predicting the mean, and on the BIEN plant data it changed nothing (docs/dev-log/discrete-lambda/README.md). Set safety_floor = TRUE to restore the old behaviour.

phylo_signal_gate
: logical. Default FALSE (it was TRUE from v0.9.1.9003 until 0.11.0.9002; same reasons as safety_floor; set TRUE to restore). When TRUE, compute per-trait Pagel's λ on training-observed cells before fitting; for traits with lambda < phylo_signal_threshold, force (r_cal_bm = 0, r_cal_gnn = 0, r_cal_mean = 1) directly and skip BM + GNN training on those traits. Requires the phytools package. Falls back to safety-floor-only behaviour (phylo_signal_gate = FALSE effective) when phytools is absent.

phylo_signal_threshold
: numeric, default 0.2. Traits with Pagel's λ below this value are routed to the grand-mean corner of the safety-floor simplex.

phylo_signal_method
: character, currently only "lambda" is fully implemented. Reserved "blomberg_k" path returns Blomberg's K via phytools::phylosig() but uses the same threshold — which is NOT dimensionally comparable; users selecting K must supply a K-appropriate threshold.

min_val_cells
: integer. Warn at fit time if any trait has fewer than min_val_cells validation cells available for gate calibration and conformal-score estimation. Default 20: with fewer than 19 cells the split-conformal level $n/(n+1)$ is below 0.95; with 38 or fewer cells the conformal score is the largest validation residual; and with very few cells gate calibration becomes essentially a coin flip between 0 and gate_cap. Recommended operational target is n_val >= 20-30 per trait; achieve this by increasing missing_frac or collecting more species. See **Calibration at small n** below.

lambda_mode
: character. Pagel-lambda mode for the BM baseline. "estimate" (default) fits a per-trait Pagel's lambda on each continuous-family (BM-eligible) latent column – continuous, count, proportion, and zi_count magnitude, via the joint MVN / threshold-joint baseline's own lambda_cols machinery when that joint path fires, or a per-column re-fit otherwise. Binary, ordinal and categorical traits

get their own lambda by default (`discrete_lambda = "estimate"`); the rest of this paragraph describes `discrete_lambda = "fixed_1"`, where discrete traits (binary, categorical, zi gate) AND ordinal have no discrete-trait analogue of Pagel's lambda of their own – there is no lambda_k to estimate for them. Under `predict_method = "per_column"` (see below) they stay fixed at lambda = 1 in every path, matching pre-S3 behaviour. Under `predict_method = "exact"` (and for traits the default `"auto"` routes to `"exact"`) they instead share the joint fit's `lambda_block` – the exact conditional's covariance model uses ONE shared phylogenetic correlation matrix `R(lambda_block)` for every column, so the Sigma estimate feeding it must itself come from an internally consistent init, not a mix of `R(1)` for discrete columns and `R(lambda_k)` for continuous ones (Shinichi's decision, docs/dev-log/exact-default/S3-default-report.md). `"fixed_1"` preserves the pre-lambda Brownian correlation matrix everywhere; `"cv"` and `"bayes"` are alternative per-column estimators that force continuous-family columns onto the per-column BM path (no joint analogue). When `predict_method = "exact"` or `joint_refine_iter > 0`, the joint (multi-trait) prediction path additionally uses the single shared `lambda_block` for cross-trait computations that need one common phylogenetic correlation matrix, never overriding a continuous-family column's own estimated lambda_k. Passed to `fit_baseline` and stored in the fitted model config. When `covariates` are supplied, the covariate-aware BM path (`bm_impute_col_with_cov()`) accepts a numeric lambda or `"estimate"`, so `lambda_mode %in% c("estimate", "fixed_1")` reaches it and each covariate-aware BM-eligible column gets its own estimated lambda; it does NOT accept `"cv"` / `"bayes"`, which fall back to lambda = 1 for BM-eligible columns with a warning from `fit_baseline`.

`discrete_lambda`

character, `"estimate"` (default) or `"fixed_1"`. With `"estimate"`, the binary, ordinal and zero-inflated-count gate liability columns of the threshold-joint baseline, and the one-vs-rest column of each categorical class, get their own Pagel's lambda, so a weak-signal discrete trait shrinks toward its prevalence instead of copying close relatives. `"fixed_1"` restores the previous behaviour (binary and ordinal at lambda = 1, or at the shared joint lambda on the "exact" route; categorical at 1). Evidence (simulated Brownian-motion and OU data, n = 100 to 1000, 30\ missing; `docs/dev-log/discrete-lambda/README.md`): estimating discrete lambda alone added about 0.01 to 0.04 discrete accuracy at Pagel's lambda of 0.3 or below; together with `safety_floor = FALSE` and `phylo_signal_gate = FALSE` (also the defaults) the gain was 0.04 to 0.12, and binary and categorical accuracy matched or beat BACE's. Costs: with lambda near 1 and n >= 1000, accuracy -0.002 to -0.005 and Brier score +0.010 to +0.016; on AVONET300's categorical traits, Brier +0.014 to +0.020 with accuracy unchanged or better. Ordinal accuracy still trails BACE's at lambda = 0.3. The categorical one-vs-rest lambdas are not reported in `$lambda_per_trait`. Passed to `fit_baseline` and stored in the fitted model config.

`joint_solver`

character. Which solver estimates the joint MVN / threshold-joint / OVR categorical baselines. `"inhouse"` (default) is byte-identical to prior releases; `"rphylopars"` delegates to `Rphylopars::phylopars()`'s converged REML fit, with automatic fallback to `"inhouse"` on failure. Passed to `fit_baseline` and stored in the fitted model config. See `docs/dev-log/2026-08-16-continuous-gap-diagnosis.md`.

`predict_method`

character. Prediction route for the in-house joint solver, passed to `fit_baseline`. `"auto"` (default, S5b) fits the baseline with both the `"exact"` and `"per_column"` routes and picks, per trait, whichever has the lower loss on that trait's validation cells (mean squared error on the z-scored latent scale for continuous/count/proportion/ordinal/zi_coun

magnitude; mean log-loss for binary/zi_count gate; mean multinomial log-loss for categorical), falling back to "exact" on a tie or when fewer than 5 validation cells belong to that trait (including when there is no validation split at all). "exact" uses the full cross-trait conditional mean and variance of vec(L) ~ MVN(0, Sigma %x% R(lambda_block)) (Hadfield & Nakagawa, 2010 sparse precision form), each column GLS-mean-centred at lambda_block before the solve; discrete liability columns share lambda_block under this route (see lambda_mode above). Falls back to "per_column" above roughly 20000 unknown cells (roughly 4000 species at 5 traits), on a singular/unusable Sigma, or when fewer than 2 joint columns or no Henderson sparse precision are available; prints a one-time message() per R session when predict_method was left at its default, or a warning() every time when "exact" was requested explicitly. "per_column" retains the original per-column conditional prediction route (no cross-trait borrowing in the prediction step). None of the three options change covariance estimation or the "rphylopars" solver. The route actually used is recorded in $model_config$predict_method_used ("exact", "per_column", or "auto") and, per trait, in $model_config$predict_method_by_trait. Under "auto" with real validation cells, $model_config $route_val_n / $score_val_n record, per trait, how many of its validation cells chose the route versus were reserved for gate calibration and conformal scoring (see fit_baseline's predict_method docs); both are NULL under an explicit "exact"/"per_column" request.

joint_refine_iter

integer, default 0L. Enables cross-trait refinement of the joint baseline's cell imputations using the estimated Sigma (the in-house solver's max_iter EM cell-refinement; R/joint_mvn_solver.R). 0L preserves current behaviour byte-for-byte. The refinement is guarded: the Sigma step must shrink each iteration, or the loop rolls back to the last good iterate and sets $diverged. Assess this opt-in control with held-out evaluation on the intended data. Has no effect on any trait predicted by the "exact" route (see fit_baseline's predict_method docs); under the default predict_method = "auto" it therefore only applies to traits "auto" routes to "per_column".

verbose — logical. Print training progress (default TRUE).

seed — optional integer. When supplied, makes stochastic training and calibration reproducible; the default NULL uses the current RNG stream.

Details

Blend formulation: The prediction is $\hat{x} = (1 - r)\mu + r\delta$, where μ is the BM baseline, δ is the model's direct prediction, and $r = \sigma(\rho) \times \text{cap}$ is a per-column learnable gate bounded in $(0, \text{gate_cap})$. When $r = 0$, the prediction collapses to the baseline. The gate is regularised toward zero via the shrinkage penalty on $\delta - \mu$, so the model defaults to the baseline unless the GNN's correction demonstrably helps on the validation set.

Training objective (per epoch):

1. A random subset of observed cells is corrupted with a learnable mask token.
2. The model predicts δ from graph context.
3. Loss = type-specific reconstruction on corrupted cells + lambda_shrink * MSE($\delta - \mu$) + lambda_gate * MSE(r).

The gate penalty on r is necessary because when $\delta = \mu$ (the BM-optimal solution for observed cells), the reconstruction and shrinkage losses both equal zero regardless of r, leaving no gradient

to close the gate. The explicit penalty ensures gates default toward zero when the GNN correction provides no benefit.

Type-specific losses:

continuous/count/ordinal MSE

binary BCE with logits

categorical cross-entropy over K latent columns

Value

An object of class `"pigauto_fit"`.

Calibration at small n

Conformal bounds use the empirical `(1 - alpha)` quantile of $|y - \hat{y}|$ on held-out validation cells. They are nominal held-out diagnostics and do not certify package-wide coverage.

Small validation sets make both the residual quantile and gate selection unstable. The `min_val_cells` warning identifies traits where this is especially relevant. `conformal_split_val = TRUE` can separate gate selection from residual estimation for eligible columns, but it does not remove the small-sample limitation. Use `cross_validate()` on the intended data and report the resulting held-out diagnostics rather than treating an interval target as certified.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
data <- preprocess_traits(traits, tree)
splits <- make_missing_splits(data$X_scaled, trait_map = data$trait_map,
                              seed = 1)
fit <- fit_pigauto(data, tree, splits = splits, epochs = 5L,
                   verbose = FALSE, seed = 1)
```

---

impute *Impute missing phylogenetic traits (convenience wrapper)*

Description

One-call interface for preprocessing, phylogenetic baseline fitting, and prediction. The default `gnn = FALSE` uses the baseline alone. Set `gnn = TRUE` to also train a GNN and calibrate its contribution. For fine-grained control, use the individual functions (`preprocess_traits`, `fit_baseline`, `fit_pigauto`, etc.) directly.

Usage

```
impute(
  traits,
  tree,
  species_col = NULL,
  trait_types = NULL,
  multi_proportion_groups = NULL,
  k_eigen = "auto",
  log_transform = TRUE,
  missing_frac = 0.25,
  n_imputations = 1L,
  covariates = NULL,
  epochs = 2000L,
  verbose = TRUE,
  seed = NULL,
  multi_obs_aggregation = c("hard", "soft"),
  lambda_mode = c("estimate", "fixed_1", "cv", "bayes"),
  discrete_lambda = c("estimate", "fixed_1"),
  joint_solver = c("inhouse", "rphylopars"),
  predict_method = c("auto", "exact", "per_column"),
  joint_refine_iter = 0L,
  em_iterations = 0L,
  em_tol = 0.001,
  em_offdiag = FALSE,
  pool_method = c("median", "mean", "mode"),
  clamp_outliers = FALSE,
  clamp_factor = 5,
  match_observed = c("none", "pmm"),
  pmm_K = 5L,
  safety_floor = FALSE,
  phylo_signal_gate = FALSE,
  phylo_signal_threshold = 0.2,
  phylo_signal_method = "lambda",
  conformal_split_val = FALSE,
  gnn = FALSE,
  ...
)
```

Arguments

traits	data.frame with species as rownames and trait columns, or (when `species_col` is supplied) a data.frame with a species column that may have multiple rows per species. Supported column types: `numeric` (continuous), `integer` (count), `factor` (binary/categorical), `ordered` (ordinal), `character` (factor → binary/categorical), `logical` (binary). See the **Trait type auto-detection** section below.
tree	object of class `"phylo"`.
species_col	character. Name of the column in `traits` that identifies species. When supplied, multiple observations per species are supported. Default `NULL` uses row names (one row per species).
trait_types	named character vector overriding the auto-detected type for specific trait columns, e.g. `c(Survival = "proportion", Parasites = "zi_count")`. Required for

the two types that cannot be inferred from R class (see **Trait type auto-detection** below). Default `NULL` (auto).

`multi_proportion_groups`
: named list declaring compositional (`multi_proportion`) traits, e.g. `list(colour = c("black", "blue", "red", "yellow"))`. Each list element names a group and gives the K trait columns that form a simplex (rows summing to 1). Encoded via CLR + per-component z-score. Multi_proportion traits *cannot* be declared through `trait_types` — use this argument instead. Default `NULL` (no multi_proportion groups).

`k_eigen`
: integer or `"auto"`. Number of spectral node features used to build the phylogenetic graph. The default `"auto"` scales the dimension to tree size; an integer overrides that choice.

`log_transform`
: logical. Log-transform every continuous trait whose observed values are all positive (default `TRUE`), then back-transform the imputations. Set it to `FALSE` when traits are already on a log scale (for example log body mass): otherwise they are logged a second time and imputed on a log(log) scale, which can bias analyses done on the log scale. With a mix of raw and already-logged traits, log the raw ones yourself and set `FALSE`. In a real-data check (PanTHERIA, 4,027 mammal species, four already-logged traits, posterior draws), the default shifted Rubin-pooled regression slopes by up to 3.9 standard errors of the complete-data slope, in a direction that depended on which values were missing; with `FALSE` the largest shift was 1.8.

`missing_frac`
: numeric. Fraction of observed cells held out for validation/test evaluation (default `0.25`). Must be in `[0, 1)`. `0` adds no extra missingness (no val/test holdout).

`n_imputations`
: integer. Number of MC-dropout imputation sets (default `1`). Must be a positive integer. Values > 1 enable between-imputation uncertainty.

`covariates`
: data.frame or matrix of environmental covariates (fully observed — no NAs). Covariates are conditioners: they inform imputation but are not themselves imputed. Numeric/integer columns are z-scored; factor/ordered columns are one-hot encoded automatically. If a variable has missing values, include it in `traits` instead. Rows are aligned to the same species / observation order as `traits` after tree matching (single-obs: rownames vs tip labels; multi-obs: `species_col` if present, otherwise input-row order). Matching `nrow` alone does not pair species to environment. Default `NULL` (no covariates).

`epochs`
: integer. Maximum GNN training epochs (default `2000`). Must be a positive integer.

`verbose`
: logical. Print progress (default `TRUE`).

`seed`
: optional integer. When supplied, makes the stochastic fitting and prediction steps reproducible; the default `NULL` uses the current RNG stream.

`multi_obs_aggregation`
: character. How to aggregate multiple observations per species before the Level-C baseline. `"hard"` (default) thresholds binary proportions at 0.5 and uses argmax for categorical. `"soft"` preserves species-level proportions and dispatches a soft E-step so that intermediate class frequencies contribute fractional liability evidence. Passed to `fit_baseline`.

`lambda_mode`
: character. Pagel-lambda mode for the BM baseline. `"estimate"` (default) fits a per-trait Pagel's lambda on each continuous-family (BM-eligible) latent column – continuous, count, proportion, and zi_count magnitude, via the joint MVN

/ threshold-joint baseline's own lambda_cols machinery when that joint path fires, or a per-column re-fit otherwise. Binary, ordinal and categorical traits get their own lambda by default (discrete_lambda = "estimate"); the rest of this paragraph describes discrete_lambda = "fixed_1", where discrete traits (binary, categorical, zi gate) AND ordinal have no discrete-trait analogue of Pagel's lambda of their own – there is no lambda_k to estimate for them. Under predict_method = "per_column" (see below) they stay fixed at lambda = 1 in every path, matching pre-S3 behaviour. Under predict_method = "exact" (and for traits the default "auto" routes to "exact") they instead share the joint fit's lambda_block – the exact conditional's covariance model uses ONE shared phylogenetic correlation matrix R(lambda_block) for every column, so the Sigma estimate feeding it must itself come from an internally consistent init, not a mix of R(1) for discrete columns and R(lambda_k) for continuous ones (Shinichi's decision, docs/dev-log/exact-default/S3-default-report.md). "fixed_1" preserves the pre-lambda Brownian correlation matrix everywhere; "cv" and "bayes" are alternative per-column estimators that force continuous-family columns onto the per-column BM path (no joint analogue). When predict_method = "exact" or joint_refine_iter > 0, the joint (multi-trait) prediction path additionally uses the single shared lambda_block for cross-trait computations that need one common phylogenetic correlation matrix, never overriding a continuous-family column's own estimated lambda_k. Passed to fit_baseline and stored in the fitted model config. When covariates are supplied, the covariate-aware BM path (bm_impute_col_with_cov()) accepts a numeric lambda or "estimate", so lambda_mode %in% c("estimate", "fixed_1") reaches it and each covariate-aware BM-eligible column gets its own estimated lambda; it does NOT accept "cv" / "bayes", which fall back to lambda = 1 for BM-eligible columns with a warning from fit_baseline.

discrete_lambda

character, "estimate" (default) or "fixed_1". With "estimate", the binary, ordinal and zero-inflated-count gate liability columns of the threshold-joint baseline, and the one-vs-rest column of each categorical class, get their own Pagel's lambda, so a weak-signal discrete trait shrinks toward its prevalence instead of copying close relatives. "fixed_1" restores the previous behaviour (binary and ordinal at lambda = 1, or at the shared joint lambda on the "exact" route; categorical at 1). Evidence (simulated Brownian-motion and OU data, n = 100 to 1000, 30\ missing; docs/dev-log/discrete-lambda/README.md): estimating discrete lambda alone added about 0.01 to 0.04 discrete accuracy at Pagel's lambda of 0.3 or below; together with safety_floor = FALSE and phylo_signal_gate = FALSE (also the defaults) the gain was 0.04 to 0.12, and binary and categorical accuracy matched or beat BACE's. Costs: with lambda near 1 and n >= 1000, accuracy -0.002 to -0.005 and Brier score +0.010 to +0.016; on AVONET300's categorical traits, Brier +0.014 to +0.020 with accuracy unchanged or better. Ordinal accuracy still trails BACE's at lambda = 0.3. The categorical one-vs-rest lambdas are not reported in $lambda_per_trait. See fit_pigauto().

joint_solver

character. Which solver estimates the joint MVN / threshold-joint / OVR categorical baselines. "inhouse" (default) is byte-identical to prior releases; "rphylopars" delegates to Rphylopars::phylopars()'s converged REML fit, with automatic fallback to "inhouse" on failure. Passed to fit_baseline and stored in the fitted model config. See docs/dev-log/2026-08-16-continuous-gap-diagnosis.md.

predict_method

character. Prediction route for the in-house joint solver, passed to fit_baseline. "auto" (default, S5b) fits the baseline with both the "exact" and "per_column" routes and picks, per trait, whichever has the lower loss on that trait's validation

cells (mean squared error on the z-scored latent scale for continuous/count/proportion/ordinal/zi_coun magnitude; mean log-loss for binary/zi_count gate; mean multinomial log-loss for categorical), falling back to "exact" on a tie or when fewer than 5 validation cells belong to that trait (including when there is no validation split at all). "exact" uses the full cross-trait conditional mean and variance of vec(L) ~ MVN(0, Sigma %x% R(lambda_block)) (Hadfield & Nakagawa, 2010 sparse precision form), each column GLS-mean-centred at lambda_block before the solve; discrete liability columns share lambda_block under this route (see lambda_mode above). Falls back to "per_column" above roughly 20000 unknown cells (roughly 4000 species at 5 traits), on a singular/unusable Sigma, or when fewer than 2 joint columns or no Henderson sparse precision are available; prints a one-time message() per R session when predict_method was left at its default, or a warning() every time when "exact" was requested explicitly. "per_column" retains the original per-column conditional prediction route (no cross-trait borrowing in the prediction step). None of the three options change covariance estimation or the "rphylopars" solver. The route actually used is recorded in fit$model_config$predict_method_used ("exact", "per_column", or "auto") and, per trait, in fit$model_config$predict_method_by_trait. Under "auto" with real validation cells, fit$model_config $route_val_n / $score_val_n record, per trait, how many of its validation cells chose the route versus were reserved for gate calibration and conformal scoring (see fit_baseline's predict_method docs).

joint_refine_iter

integer, default 0L. Enables cross-trait refinement of the joint baseline's cell imputations using the estimated Sigma (the in-house solver's max_iter EM cell-refinement; R/joint_mvn_solver.R). 0L preserves current behaviour byte-for-byte. The refinement is guarded: the Sigma step must shrink each iteration, or the loop rolls back to the last good iterate and sets $diverged. Assess this opt-in control with held-out evaluation on the intended data. Has no effect on any trait predicted by the "exact" route (see fit_baseline's predict_method docs); under the default predict_method = "auto" it therefore only applies to traits "auto" routes to "per_column". Passed to fit_baseline and stored in the fitted model config.

em_iterations

integer. Phase 6 EM iterations for the threshold-joint baseline (binary + ordinal + OVR categorical). Default 0L preserves v0.9.1 behaviour byte-for-byte. When >= 2L, the BM rate Σ learned by the joint solver (joint_solver; the in-house solver by default) at iteration k is fed back as the per-trait prior SD at iteration $k + 1$, up to em_iterations times or until em_tol convergence. Passed to fit_baseline.

em_tol

numeric. Relative-Frobenius convergence tolerance for the Phase 6 / 7 EM loop. Default 1e-3.

em_offdiag

logical. Phase 7 opt-in: when TRUE AND em_iterations >= 2L, each liability cell's prior uses the full conditional-MVN from Σ's off-diagonal entries, so that observing one discrete trait shifts (not just tightens) the prior on correlated other traits. Binary + ordinal only; OVR categorical stays on Phase 6 diagonal. Default FALSE. Passed to fit_baseline.

pool_method

character. How to pool multiple imputation draws (n_imputations > 1) for count, proportion, and zi_count magnitude traits: "median" (default) takes the per-cell median of the M decoded draws — robust to dropout-noisy latents amplified by expm1() / plogis() decoders. "mean" restores the pre-v0.9.2 arithmetic-mean pooling. "mode" (Phase H, v0.9.1.9010+) is intended for ordinal traits:

per-cell majority vote across the M draws, avoiding the integer-mean-round bias toward middle classes. For continuous-family traits, "mode" falls back to "median". Binary / categorical / multi_proportion traits always pool by probability average; unaffected by this argument. See issue #40.

clamp_outliers logical. Phase G (v0.9.1.9011+). When TRUE, post-back-transform predictions for log-transformed continuous, count, and zi_count magnitude traits are capped at tm$obs_max * clamp_factor (and tm$obs_max is the observed maximum on the original scale, recorded at preprocess time). Targets the AVONET Mass tail-extrapolation mode documented in useful/MEMO_2026-05-01_avonet_mass_diag.md where a $+3$-4σ latent overshoot becomes a 50x-100x value error after expm1(). Default FALSE preserves v0.9.1 behaviour exactly.

clamp_factor numeric scalar (>= 1). Multiplicative factor on the observed maximum used by clamp_outliers. Default 5 (Tukey-style outlier definition: anything >= 5x the observed max is implausible). Ignored when clamp_outliers = FALSE.

match_observed character, one of c("none", "pmm"). Phase G' (v0.9.1.9012+). Pass-through to predict.pigauto_fit. When "pmm", uses Predictive Mean Matching for log-transformed continuous, count, zi_count magnitude, and proportion traits: imputed values are drawn from the observed value pool, never extrapolated.

When to use: PMM is a niche feature. pigauto already provides conformal prediction intervals calibrated against held-out residuals and stochastic prediction draws for diagnostics. PMM is only worth enabling for: (a) methodological comparison against mice, or (b) workflows that specifically require imputed values to come from the observed data pool. For tail safety, prefer clamp_outliers = TRUE. PMM failed the downstream fixed-effect redesign pilot and is unsupported for inference; see multi_impute_analysis() for the analysis-aware backend.

Default "none" preserves pre-G' behaviour.

pmm_K integer (>= 1). Donor pool size for PMM. Default 5L (mice convention). Ignored when match_observed = "none".

safety_floor logical. Default FALSE (it was TRUE from v0.9.1.9002 until 0.11.0.9002; see fit_pigauto() for why it changed; set TRUE to restore). When TRUE, calibration searches the 3-way simplex r_BM * BM + r_GNN * GNN + r_MEAN * MEAN so the grand mean is always in the candidate set. Under the validation metric used for calibration, the selected candidate cannot be worse than that grand-mean corner on the validation cells. This is a validation safeguard, not a guarantee about future held-out data. When FALSE, the v0.9.1 1-D calibration is used exactly. See the Safety floor section below.

phylo_signal_gate, phylo_signal_threshold, phylo_signal_method

Pass-through to fit_pigauto(). See that help page for details. phylo_signal_gate now defaults to FALSE (it was TRUE from v0.9.1.9003 until 0.11.0.9002): with Pagel's λ estimated for every trait type the gate no longer improved accuracy in any tested setting, and at λ of 0 to 0.1 it was worse than predicting the mean. Set phylo_signal_gate = TRUE to restore the old behaviour.

conformal_split_val

logical. Default FALSE. When TRUE, the validation cells are split 50/50 per trait into a gate-calibration half and a conformal-scoring half, so the conformal quantile is not post-selected on the same cells the gate was tuned on. Costs interval width at small validation sizes; see fit_pigauto for details. Previously reachable only via ...; exposed here for discoverability.

gnn logical. When TRUE, trains the attention-based GNN correction (see fit_pigauto). When FALSE (default), no GNN is constructed or trained – impute() makes no

torch/GPU calls, and the fit is the phylogenetic baseline alone (optionally reweighted against a grand-mean floor when `safety_floor = TRUE`). With the defaults (`safety_floor = FALSE, phylo_signal_gate = FALSE`) this is the pure traditional-stats arm. Production predictions (`result$completed`, `result$prediction`) use a tax-free `baseline_full` fit on ALL observed cells (no val/test hold-out); `result$evaluation` and every other scorer keep using the held-out `baseline`, so evaluation never leaks test cells. User `covariates` enter pigauto only through the GNN, so under `gnn = FALSE` they are ignored (with a warning); without a validation split (`missing_frac = 0`) the fit is pure baseline and carries no conformal scores. Default `FALSE` since pigauto 0.11.0.9001: on simulated and real data at the current baseline defaults the GNN did not lower imputation error and took 20 to 90 times longer to fit (`docs/dev-log/arc/2026-10-02-campaign-gnn-rerun.md`). Set `gnn = TRUE` to train it, for example to use `covariates`, which only the GNN uses, or to restore the behaviour of earlier versions.

... additional arguments passed to `fit_pigauto`.

Value

An object of class `"pigauto_result"` with components:

completed The input `traits` data.frame with observed values preserved and only missing cells filled in. This is the primary output – typically what users want.

imputed_mask Logical matrix (same shape as `completed`) that is `TRUE` for cells that were imputed (originally `NA`) and `FALSE` for observed cells.

prediction A `pigauto_pred` object from `predict.pigauto_fit` containing raw model predictions for every cell (observed + missing), per-cell uncertainty (`$se` – see `predict.pigauto_fit`'s "What `$se` means per trait type" section; it is **not** uniformly a standard error), class probabilities, and conformal intervals.

fit The trained `pigauto_fit` object.

baseline The phylogenetic baseline.

data The preprocessed `pigauto_data` object.

splits The val/test splits (or `NULL` if `missing_frac = 0`).

evaluation Evaluation metrics on test set (or `NULL`).

check A compact `pigauto_check` preflight summary.

Trait type auto-detection

pigauto infers each trait's type from its R class — no `trait_types` argument is needed for most data:

R class	pigauto type
`numeric`	continuous (auto-log if all-positive)
`integer`	count
`factor` with 2 levels	binary
`factor` (unordered) with >2 levels	categorical
`ordered` / `factor(..., ordered = TRUE)`	ordinal
`character`	→ factor → binary or categorical
`logical`	binary

Two types cannot be inferred from class alone and *must* be declared via `trait_types`:

`"proportion"` A `numeric` bounded 0–1, e.g. survival rate: `trait_types = c(Survival = "proportion")`.

`"zi_count"` An `integer` with excess zeros, e.g. parasite count: `trait_types = c(Parasites = "zi_count")`.

Use the `trait_types` argument directly (it is an explicit parameter, not a `...` pass-through).

Traits vs covariates

The distinction is **functional, not ontological**: a trait is something you want to impute (NA values allowed in `traits`); a covariate is something you use to sharpen imputation accuracy (must be fully observed, passed via `covariates`). The same variable can be either depending on the scientific question.

Examples:

- **IUCN status with Data Deficient species** → put it in `traits` as `ordered(c("LC","NT","VU","EN","CR"))` so pigauto predicts the unknown categories.
- **IUCN status fully known for all species** → pass as a covariate to inform imputation of other traits (e.g. body mass, range size).
- **Realm / biome (factor)** → pass as a covariate; pigauto one-hot encodes factor columns automatically (v0.6.1+).

Variables that belong in `traits`: anything with missing values you care about predicting. Variables that belong in `covariates`: fully observed, exogenous to the trait space (geography, climate, habitat, experimental treatment).

Safety floor (v0.9.1.9002+)

With `safety_floor = TRUE` (opt-in since 0.11.0.9002; it was the default from v0.9.1.9002), the post-training calibration grid searches a 3-way convex combination of the Brownian-motion baseline, the GNN delta, and the per-trait grand mean. The simplex is sampled at step 0.05 (231 candidates per latent column). Because the corner `(0, 0, 1)` — pure grand mean — is always in the grid, the selected candidate cannot be worse than the grand-mean corner on the validation cells under the calibration metric. The fit object gains four new slots: `r_cal_bm`, `r_cal_gnn`, `r_cal_mean` (each a named numeric of length `p_latent`), and `mean_baseline_per_col`.

Set `safety_floor = FALSE` to reproduce the pre-v0.9.1.9002 1-D calibration bit-identically (no mean term; `r_cal_mean = 0`; `r_cal_bm = 1 - r_cal_gnn`). See `specs/2026-04-23-safety-floor-mean-gate-desig` for the design rationale and `plans/2026-04-23-safety-floor-mean-gate.md` for the implementation plan.

What gets imputed (read this first)

pigauto only *imputes* cells that are `NA` in the input. Observed cells are preserved as-is in `result$completed`. The slot `result$prediction$imputed` contains the model's prediction for **every** cell – observed and missing alike – and is intended for diagnostics (e.g. checking calibration on training cells), *not* as the imputed-values output. The imputed values themselves are `result$completed[result$imputed_mask]`.

Common pitfall. If you call `impute()` on a fully observed trait matrix (no `NA`s among matched target rows), there is nothing to impute. `impute()` stops before graph construction or training and directs you to `cross_validate()` for predictive assessment. To exercise imputation, supply a real missing target cell.

Imbalanced K-class traits. At default settings (`n_imputations = 1L`, `pool_method = "median"`), a small ordinal / categorical trait whose marginal distribution is heavily skewed (e.g. AVONET `Migration` is ~78\ ~14\ collapse onto a corner that predicts the majority class everywhere. Compare

candidate settings, including `pool_method = "mode"`, with `cross_validate()` on the intended trait table rather than applying a generic setting.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
df <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                c("Mass", "Wing.Length"), drop = FALSE]
rownames(df) <- tree$tip.label
df$Mass[seq_len(3L)] <- NA_real_
result <- impute(df, tree, epochs = 5L, verbose = FALSE)
head(result$completed)
```

---

load_pigauto *Load a saved pigauto model*

Description

Reads a `pigauto_fit` object previously saved with `save_pigauto`.

Usage

```
load_pigauto(path)
```

Arguments

`path`	character. File path to load from.

Value

An object of class `"pigauto_fit"`.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
traits$Mass[seq_len(3L)] <- NA_real_
fit <- impute(traits, tree, epochs = 5L, verbose = FALSE)$fit
path <- tempfile(fileext = ".pigauto")
save_pigauto(fit, path)
pred <- predict(load_pigauto(path))
```

`make_missing_splits` *Split cells into train/val/test for imputation evaluation*

Description

Randomly designates a fraction of cells as "missing" and splits them into validation and test sets. When a `trait_map` is supplied, masking operates at the **original trait level** – all latent columns belonging to one trait are held out together (important for categorical traits).

Usage

```
make_missing_splits(
  X,
  missing_frac = 0.25,
  val_frac = 0.25,
  seed = NULL,
  trait_map = NULL,
  mechanism = c("MCAR", "MAR_trait", "MAR_phylo", "MNAR"),
  mechanism_args = list(),
  tree = NULL
)
```

Arguments

`X`	numeric matrix (species x latent columns from `preprocess_traits`). Used only for dimensions.
`missing_frac`	numeric. Fraction of all (species, trait) cells to designate as missing (default `0.25`).
`val_frac`	numeric. Fraction of missing cells for validation (default `0.25`); the rest become the test set.
`seed`	optional integer. When supplied, makes split construction reproducible; the default `NULL` uses the current RNG stream.
`trait_map`	list of trait descriptors (from `pigauto_data`). If `NULL`, masking is applied per latent column (v0.1 behaviour).
`mechanism`	character. Missingness mechanism: `"MCAR"` (default, uniform random), `"MAR_trait"` (trait-dependent), `"MAR_phylo"` (clade-structured), or `"MNAR"` (value-dependent).
`mechanism_args`	named list of mechanism-specific parameters: **For** `"MAR_trait"`**:** `driver_col` (integer, column index in `X` that drives missingness; default 1), `beta` (numeric, severity; default 2.0). **For** `"MAR_phylo"`**:** `n_clades` (integer, number of high-missingness clades; default 2), `p_clade` (numeric, within-clade missingness probability; default 0.7), `p_base` (numeric, background missingness probability; default 0.1). **For** `"MNAR"`**:** `beta` (numeric, severity; default 2.0).
`tree`	object of class `"phylo"`. Required for `mechanism = "MAR_phylo"`, ignored otherwise.

Details

The returned index vectors use linear (column-major) indexing. Both original-trait-space and latent-space indices are returned when a `trait_map` is present.

Value

A list with:

val_idx Integer vector of linear indices (latent space).

test_idx Integer vector of linear indices (latent space).

val_idx_trait Integer vector in original-trait space (if `trait_map` supplied).

test_idx_trait Integer vector in original-trait space (if `trait_map` supplied).

n Number of species (rows).

p Number of latent columns.

n_traits Number of original traits.

mask Logical matrix (n x p_latent). `TRUE` = observed.

mechanism Character string of the mechanism used.

Examples

```
X <- matrix(rnorm(100), nrow = 20)
splits <- make_missing_splits(X, missing_frac = 0.25, seed = 1)
length(splits$val_idx)

# MAR: missingness depends on another trait
splits_mar <- make_missing_splits(X, mechanism = "MAR_trait",
  mechanism_args = list(driver_col = 1, beta = 2))
```

---

mask_missing *Create an observed/missing mask matrix*

Description

Returns a logical matrix of the same dimensions as `X`, with `TRUE` where values are observed (not `NA`).

Usage

```
mask_missing(X)
```

Arguments

`X`	numeric matrix (species x traits or species x latent columns).

Value

Logical matrix, `TRUE` = observed.

Examples

```
X <- matrix(c(1, NA, 3, 4), nrow = 2)
mask_missing(X)
```

`multi_impute` *Generate experimental stochastic completion datasets*

Description

Run pigauto's full imputation pipeline and return `M` stochastic completions of the trait matrix instead of a single point estimate. The conformal-width and Brownian/MC-dropout draws returned here are experimental prediction-diagnostic draws. Do not use these datasets for downstream inference or Rubin pooling. For continuous traits, `draws_method = "posterior"` instead returns proper Bayesian posterior imputations for the analyses described in "Posterior draws" below: fit the analysis to them with `with_imputations()` and pool the resulting fits with `pool_mi()`. The separate analysis-aware backend, `multi_impute_analysis()`, covers its own documented narrow regime.

Usage

```
multi_impute(
  traits,
  tree,
  m = 100L,
  draws_method = c("auto", "conformal", "mc_dropout", "posterior"),
  species_col = NULL,
  trait_types = NULL,
  multi_proportion_groups = NULL,
  log_transform = TRUE,
  missing_frac = 0.25,
  covariates = NULL,
  epochs = 2000L,
  verbose = TRUE,
  seed = NULL,
  gnn = FALSE,
  lambda_mode = c("estimate", "fixed_1", "cv", "bayes"),
  posterior_control = list(),
  ...
)
```

Arguments

`traits`	data.frame with species as rownames and trait columns. Same input format as `impute()`. Supported column types are numeric, integer, factor, ordered factor, and logical.
`tree`	object of class `phylo` aligned with `traits`.
`m`	integer. Number of stochastic completion datasets to generate (default `100`). Observed cells are identical across all `M` datasets; only originally-missing cells vary.
`draws_method`	character. How stochastic draws are generated for missing cells. One of:
	`"auto"` (default; previously `"conformal"`) Use `"posterior"` when the data fit its requirements (every trait continuous, one row per species, no `covariates`, no `multi_proportion_groups`); otherwise use `"conformal"` and print a message saying why and that the draws cannot be pooled. The chosen method is stored in `result$draws_method`. To get the previous default, set `draws_method = "conformal"`.

"conformal" Use this method when "auto" falls back because the posterior sampler's requirements are not met, or when explicitly requested. Run the model once, then sample each originally-missing cell from a Normal distribution centred on the point estimate with SD = conformal_score / 1.96. Converting a split-conformal residual quantile to a Normal scale is a heuristic; a nominal held-out conformal diagnostic does not establish that these draws are proper multiple imputations. Falls back to BM-SE-based Normal sampling when conformal scores are unavailable, and to Bernoulli / Categorical draws for discrete traits.

"mc_dropout" Run M stochastic GNN forward passes in training mode (dropout active) on top of stochastic Brownian-motion baseline draws. Brownian draws still contribute between-draw variation when a calibrated GNN gate is zero.

"posterior" Continuous traits only, one row per species, no covariates. Fit a Bayesian multivariate phylogenetic mixed model by MCMC and return m completions drawn from the posterior predictive distribution of the missing cells. No GNN is fitted. See "Posterior draws" below. In simulation (continuous traits, 30% missing completely at random, 200 datasets per cell, n = 100 to 1000, lambda 0.3 to 1, trait correlation 0 or 0.5), a Rubin-pooled downstream PGLS slope and phylogenetic correlation had 95% interval coverage of 0.925 to 0.975 in every cell. At lambda = 1 with trait correlation 0.5, slope coverage was 0.955, 0.940 and 0.965 at n = 100, 300 and 1000 (proper frequentist multiple imputation: 0.955, 0.920, 0.935). At n = 100 with correlated traits the pooled slope was biased by about -0.02 towards zero, with coverage still 0.96. The previous residual prior (posterior_control = list(residual_prior = "iw")) attenuated the relationship between correlated traits at lambda near 1: coverage 0.96, 0.88 and 0.75 at the same three n.

Conformal and MC-dropout draws perturb missing cells around a point prediction rather than drawing them jointly from their conditional distribution given the observed data. In a 16-regime simulation of a downstream phylogenetic regression (PGLS slope, two traits, 120 replicates per regime), conformal draws biased the pooled slope by -0.20 to -0.46, and the pooled 95% intervals covered the truth in 0 to 17% of replicates, in all 16 regimes; MC-dropout draws biased it by -0.03 to -0.38. For downstream inference on continuous traits use "posterior".

species_col character or NULL. If set, marks the column in traits containing species identifiers and enables multiple observations per species. See impute() for details.

trait_types named character vector overriding auto-detected trait types for specific columns. Required for "proportion" and "zi_count". See impute() and preprocess_traits(). Default NULL (auto-detect).

multi_proportion_groups named list declaring compositional trait groups (rows summing to 1), e.g. list(diet = c("plant", "invert", "vert")). Forwarded to impute() / preprocess_traits(). Default NULL.

log_transform logical. Log-transform every continuous trait whose observed values are all positive (default TRUE), then back-transform the imputations. Set it to FALSE when traits are already on a log scale (for example log body mass): otherwise they are logged a second time and imputed on a log(log) scale, which can bias analyses done on the log scale. With a mix of raw and already-logged traits, log the raw ones yourself and set FALSE. In a real-data check (PanTHERIA, 4,027

mammal species, four already-logged traits, posterior draws), the default shifted Rubin-pooled regression slopes by up to 3.9 standard errors of the complete-data slope, in a direction that depended on which values were missing; with FALSE the largest shift was 1.8.

missing_frac — numeric. Fraction of observed cells held out for validation/test during training (default 0.25). Passed through to impute().

covariates — data.frame or matrix of environmental covariates (fully observed, numeric). Passed through to impute(). Default NULL (no covariates).

epochs — integer. Maximum GNN training epochs (default 2000).

verbose — logical. Print progress (default TRUE).

seed — optional integer. When supplied, makes fitting and imputation draws reproducible; the default NULL uses the current RNG stream.

gnn — logical. Passed through to impute() / fit_pigauto(). When TRUE, the usual GNN correction is trained. When FALSE (default), no GNN is used (baseline-only fit; see fit_pigauto()). With draws_method = "mc_dropout" this degrades gracefully to BM-posterior draws (there is no dropout to run) and a one-time message is printed; draws_method = "conformal" is unaffected. User covariates are ignored under gnn = FALSE (with a warning), as in impute(). The default changed from TRUE to FALSE in 0.11.0.9001; see impute().

lambda_mode — character. Pagel-lambda mode for the BM baseline, forwarded to impute() / fit_pigauto(). "estimate" (default) fits a per-trait Pagel's lambda on each continuous-family (BM-eligible) latent column. Discrete traits estimate their own lambda by default (discrete_lambda = "estimate", passed through ... to impute()); set discrete_lambda = "fixed_1" to retain lambda = 1 for their baseline. Joint exact prediction can also use a shared lambda_block for its cross-trait conditional. "fixed_1" preserves the pre-lambda Brownian correlation matrix everywhere; "cv" and "bayes" are alternative per-column estimators. See fit_pigauto() for the full contract, including the predict_method = "exact" / joint_refine_iter > 0 interaction with lambda_block.

posterior_control — list of settings for draws_method = "posterior" (ignored otherwise). Elements, with defaults:

- n_chains 4L. Number of MCMC chains. Chain 1 starts at the per-trait REML Pagel's lambda clamped to [0.02, 0.98] (up to 2,000 tips; above that, or if the REML fit fails, at lambda = 0.9), and the other chains start from values dispersed around it. mi$posterior$start holds the REML lambda and chain 1's lambda before clamping.
- n_iter 5000L. Sweeps per chain after burn-in.
- burnin 1000L. Burn-in sweeps per chain (also used to tune the Metropolis step sizes).
- thin Default: n_iter integer-divided by ceiling(keep_draws / n_chains), so that the chains together keep keep_draws sweeps.
- keep_draws 1000L. Target number of kept posterior predictive draws across chains, used for the per-cell intervals. Must be at least m. The default thin keeps at least this many whenever n_iter is at least ceiling(keep_draws / n_chains). If a user-set n_iter or thin keeps fewer (but at least m), a warning is raised and the intervals use the sweeps that were kept; keeping fewer than m is an error.

param_uncertainty "full" (default), "none" or "both". "none" is an improper plug-in mode for validation only: the covariance matrices are fixed at their posterior means from a full run, and only the missing cells (and the trait means) are drawn. Its result carries mi_workflow = "pigauto_posterior_plugin_diagno and with_imputations() and pool_mi() refuse it. "both" is also for validation only: it runs the sampler once and returns the proper results exactly as "full" does, plus the plug-in draws from the same run in mi$posterior_improper.

seed Integer or NULL. Defaults to seed.

auto_extend TRUE. If the chains fail the convergence rule (any split R-hat at least 1.05 or any bulk ESS at most 400), every chain continues from where it stopped for another n_iter sweeps and the diagnostics are recomputed on all sweeps after burn-in; this repeats until the rule is met or max_extend extensions have been made. Burn-in is not repeated and the Metropolis step sizes stay as tuned in burn-in, so an extended chain is the same chain as one run longer from the start. After k extensions the kept draws are every thin * (k + 1)-th sweep, spread over the whole run, so their number does not change. A fit that meets the rule first time is not affected. FALSE never extends.

max_extend 3L. Maximum number of extensions, a whole number from 0 to 10. The default allows at most 4 times n_iter sweeps per chain after burn-in.

residual_prior "sep" (default) or "iw". The default separation prior gives each residual standard deviation a half-Cauchy prior scaled by the trait's observed standard deviation, and gives the residual correlation a uniform (LKJ(1)) prior. This allows residual covariance to approach zero when traits are close to Brownian motion. "iw" selects the legacy inverse-Wishart prior used before pigauto 0.11.0.9002; it is retained for reproducing earlier results.

... additional arguments forwarded to fit_pigauto() via impute(). See fit_pigauto() for the full list, including the optional safety_floor control described below.

Details

With the default draws_method = "auto", eligible continuous-trait data use posterior draws; other inputs use conformal prediction-diagnostic draws. Check result$draws_method before downstream fitting.

The conformal and MC-dropout draws do not condition on a declared substantive analysis model. Consequently, stochastic variation alone does not make them proper or congenial multiple imputations. The analysis-aware backend requires the analysis model before generating draws and dispatches only across its documented supported model classes.

draws_method = "conformal": Run the model once; missing cells are sampled from $x_{ij}^{(k)} \sim \mathrm{N}(\hat{\mu}_{ij},\ (q_j/1.96)^2)$ where q_j is the trait-level split-conformal residual quantile. Dividing this quantile by 1.96 is a pragmatic Normal-scale construction, not an inference consequence of a nominal held-out conformal diagnostic. For discrete traits (binary, categorical) it uses Bernoulli / categorical draws from the estimated probability vector. For zi_count the gate is Bernoulli from P(nonzero) and the conditional magnitude is drawn on the log1p-z latent using the conformal score (or the magnitude latent SE when scores are unavailable) — not the reported SE of expected count. For multi_proportion groups it draws the K CLR latent columns with their BM latent SEs, projects back to sum-zero CLR space, and decodes to the simplex.

draws_method = "mc_dropout": Run M GNN forward passes in training mode (dropout active) on top of stochastic BM baseline draws. When r_cal = 0, the GNN-dropout term disappears but the

BM draw still contributes between-draw variance.

`draws_method = "posterior"`: see the "Posterior draws" section.

Nakagawa & Freckleton (2008, 2011) review the consequences of ignoring missing data in ecological and comparative analyses and argue for multiple imputation as the default.

Value

An object of class `"pigauto_mi"` with components:

`datasets` A list of length `m`. Each element is a data.frame with the same shape and column types as the input `traits`; observed cells are preserved and missing cells are filled with the corresponding stochastic draw. These datasets are for prediction diagnostics, not downstream inference, except with `draws_method = "posterior"` (unless `posterior_control$param_uncertainty = "none"`), whose datasets are posterior imputations: fit the analysis to them with `with_imputations()` and pool the fits with `pool_mi()` (see `posterior` below and "Posterior draws").

`m` Number of stochastic completion datasets.

`pooled_point` A single data.frame whose missing cells are replaced by the MC-averaged point estimate. Convenient for reporting but does *not* provide a valid downstream MI analysis.

`se` Matrix of per-cell uncertainty summaries combining the baseline SE and the between-draw standard deviation. A **descriptive spread for reporting and ranking only** — it is **not** a Rubin's-rules pooled standard error (it contains no within-imputation variance component and no small-sample df correction) and must not be used for downstream pooling.

`mi_workflow` `"pigauto_diagnostic_mi"`, recording that these are prediction-diagnostic completions and cannot be passed to `with_imputations()` or `pool_mi()`.

`imputed_mask` Logical matrix; `TRUE` where a cell was originally missing.

`fit` The underlying `pigauto_fit` object, retained for diagnostics and for calls to `predict()` on new data.

`data` The `pigauto_data` object.

`tree` The input phylogeny.

`species_col` Passed-through species-column name or `NULL`.

`posterior` `draws_method = "posterior"` only. A list with `cell_interval` (data.frame: `row` (row of `traits`), `trait`, `lower`, `upper`, `median`; 95% posterior predictive interval on the original scale), `diagnostics` (data.frame: `parameter`, `rhat`, `ess_bulk`, with attributes `"converged"`, `"n_extensions"` (the number of automatic extensions made) and `"sweeps_per_chain"` (the sweeps after burn-in per chain that the diagnostics use)), `params` (list: `Sigma_P` and `Sigma_E` as `K x K x draws` arrays, `lambda` and `mu` as `draws x K` matrices, on the latent scale), `converged`, `control`, `hyper` (priors), `start` (REML lambda and chain-1 start lambda, before clamping to [0.02, 0.98]), `draw_index` (kept sweeps used for the `m` datasets), `wall_s` and `sweeps` (all sweeps, burn-in included, summed over chains). For this method `fit` is `NULL`, `se` holds the posterior predictive SD of each imputed cell, the class is `c("pigauto_posterior_mi", "pigauto_mi", "list")` and `mi_workflow` is `"pigauto_posterior_mi_v1"` (`"pigauto_posterior_plugin_di` with `posterior_control$param_uncertainty = "none"`, which `with_imputations()` and `pool_mi()` refuse).

`posterior_improper` Only with `posterior_control$param_uncertainty = "both"` (validation only). A list with `datasets` (`m` completed data.frames drawn with the covariance matrices fixed at their posterior means, from the same chain run as `datasets`), `cell_interval` (same columns as `posterior$cell_interval`), and the fixed `Sigma_P` and `Sigma_E`. Not for downstream inference: it ignores parameter uncertainty.

When to use this

With `draws_method = "conformal"` or `"mc_dropout"`, this function is useful for comparing stochastic prediction behavior from one tree; those draws are not for downstream inference. With `draws_method = "posterior"` (continuous traits), it returns proper posterior imputations for the analyses described in "Posterior draws": `with_imputations()` fits the analysis to them and `pool_mi()` pools those fits. `multi_impute_analysis()` is the separate analysis-aware backend for its own documented regime.

`multi_impute_trees()` provides an experimental posterior-tree sensitivity path, but tree uncertainty is not supported by `multi_impute_analysis()`.

Posterior draws

With `draws_method = "posterior"`, the latent (z-scored, optionally log-transformed) traits follow the multivariate phylogenetic mixed model

$$\mathrm{vec}(Y) \sim \mathrm{N}(1\mu^\top,\ \Sigma_P \otimes R + \Sigma_E \otimes I_n),$$

with R the phylogenetic correlation matrix of the tree, full $K \times K$ phylogenetic (Σ_P) and residual (Σ_E) covariance matrices, and a flat prior on μ. The implied Pagel's lambda of trait k is $\Sigma_P[k,k]/(\Sigma_P[k,k] + \Sigma_E[k,k])$.

The phylogenetic covariance uses a parameter-expanded inverse-Wishart prior on its standardized covariance matrix and Normal priors on the expansion parameters (Hadfield 2010; Gelman 2006). The default residual covariance prior uses a separation strategy: residual standard deviations have half-Cauchy priors scaled by the observed latent-trait standard deviations, and the residual correlation has an LKJ(1) prior. The legacy `posterior_control$residual_prior = "iw"` option instead uses an inverse-Wishart prior on the residual covariance. Prior settings are returned in `mi$posterior$hyper`.

The sampler draws the phylogenetic effects, the trait means and the missing cells as one block with a sparse Cholesky factor of the Hadfield and Nakagawa (2010) node precision, and updates the covariance matrices by parameter-expanded Gibbs steps plus Metropolis moves with the phylogenetic effects integrated out. Convergence is checked with rank-normalised split R-hat and bulk effective sample size (Vehtari et al. 2021); `mi$posterior$diagnostics` reports them. When any R-hat is at least 1.05 or any ESS is at most 400, the chains are extended automatically (`posterior_control$auto_extend` and `max_extend`), and a warning is raised if the rule still fails after the last extension.

The `m` datasets are posterior predictive draws spaced evenly across the kept sweeps of all chains. Per-cell 95% intervals in `mi$posterior$cell_interval` come from all kept sweeps (`keep_draws` of them by default), never from the `m` datasets.

These draws come from a linear-Gaussian model of the imputed traits on the scale where they were imputed (the log scale for traits log-transformed by `log_transform`). They are proper imputations for analyses that are linear in the imputed traits on that scale and whose variables are all among the imputed traits, for example a phylogenetic regression of one imputed trait on others (on the log scale for log-transformed traits). Not covered: covariates from outside the imputed traits, because the imputation model does not contain them; nonlinear terms (such as squares) or interactions among imputed traits; and analysing a log-transformed trait on its raw scale. The GNN is not used, and `gnn`, `epochs`, `missing_frac`, `lambda_mode` and other fitting arguments are ignored (a message lists any that were supplied). Non-continuous traits, multiple observations per species (or any `species_col`), `covariates`, and input with no missing cells in the rows of `traits` (tree tips absent from `traits` do not count) are errors.

Optional safety floor

When `fit_pigauto()` is called with `safety_floor = TRUE` (opt-in; the default is `FALSE`), the 3-way blend `r_BM * BM + r_GNN * GNN + r_MEAN * MEAN` propagates through prediction-diagnostic draws through `predict.pigauto_fit()`. The posterior sampler uses its own model and does not use this blend. For `draws_method = "mc_dropout"` the mean term contributes no between-draw variance (it is a deterministic scalar per column); between-draw variance comes from the BM-draw and GNN-dropout terms only. For `draws_method = "conformal"` the blend centre is the 3-way prediction and conformal scores remain calibrated on the blended residuals.

References

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See Also

`impute()` for point imputation and `multi_impute_analysis()` for the narrow analysis-aware inferential backend.

Examples

```
library(pigauto)
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
df <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                c("Mass", "Wing.Length"), drop = FALSE]
rownames(df) <- tree$tip.label
df$Mass[seq_len(3L)] <- NA_real_
mi <- multi_impute(df, tree, m = 2L, draws_method = "conformal",
                   epochs = 5L, verbose = FALSE)
print(mi)
lapply(mi$datasets, head)
```

`multi_impute_analysis` *Analysis-aware multiple imputation for narrow regression models*

Description

Generate multiple imputations conditional on a specified downstream regression model. Validity requires MAR, a correctly specified conditional imputation model, and adequate SMCFCS or jomo convergence. This experimental interface is deliberately narrow: exactly one continuous covariate may contain missing values, while the outcome, all other predictors, auxiliary variables, and any grouping variable must be fully observed.

Usage

```
multi_impute_analysis(
  data,
  formula,
  missing,
  model = c("lm", "glm", "lmer"),
  m = 50L,
  auxiliary = character(),
  seed = NULL,
  control = list()
)
```

Arguments

`data`	A data frame containing the analysis variables.
`formula`	A two-sided additive model formula. For `model = "lmer"`, exactly one random-intercept term of the form `(1 | group)` is required.
`missing`	A single character string naming the continuous numeric covariate to impute. It must occur as a main-effect predictor in `formula` and must be the only column in `data` containing missing values.
`model`	One of `"lm"`, `"glm"`, or `"lmer"`. The `"glm"` route is restricted to binomial-logit regression. The `"lmer"` route is restricted to a Gaussian random-intercept model.
`m`	Integer number of completed datasets. Must be at least two.
`auxiliary`	Character vector naming fully observed numeric columns used only by the imputation model. They cannot duplicate formula variables. Derived terms must be created explicitly in `data` before calling this function.
`seed`	Optional integer random seed. When supplied, makes the imputation draws reproducible; the default `NULL` uses the current RNG stream.
`control`	Named list of engine controls. The `"lm"` route accepts no controls. The `"glm"` route accepts `numit` (default 20) and `rjlimit` (default 100000). The `"lmer"` route accepts `nburn` (default 1000) and `nbetween` (default 100).

Value

An object with classes "pigauto_analysis_mi" and "pigauto_mi". Its datasets component is a list of m completed data frames compatible with with_imputations() and pool_mi(). The object also records the analysis model, formula, engine, controls, engine version, seed, runtime provenance, and imputed-variable metadata. Any engine warning aborts the call instead of returning an inference-ready object. mi_workflow is a stable analysis-aware provenance marker propagated by with_imputations() and checked by pool_mi().

Experimental scope

This function does not fit pigauto's phylogenetic baseline or graph neural network and does not propagate posterior-tree uncertainty. The "lm" route uses proper Bayesian Normal-regression draws. The "glm" route requires the optional smcfcs package, and the "lmer" route requires the optional jomo and lme4 packages. Interactions, transformed terms, random slopes, nested or crossed random effects, multiple incomplete variables, missing outcomes, and pooling of variance components or BLUPs are unsupported.

See Also

multi_impute(), with_imputations(), pool_mi()

Examples

```
set.seed(11)
dat <- data.frame(
  y = stats::rnorm(80),
  x = stats::rnorm(80),
  z = stats::rnorm(80)
)
dat$x[seq(4, 80, by = 5)] <- NA_real_
mi <- multi_impute_analysis(
  dat, y ~ x + z, missing = "x", model = "lm", m = 5, seed = 12
)
fits <- with_imputations(
  mi, function(d) stats::lm(y ~ x + z, data = d), .progress = FALSE
)
pool_mi(fits)
```

---

multi_impute_trees *Posterior-tree prediction sensitivity*

Description

Run pigauto's full imputation pipeline on each of T posterior phylogenies, generating m_per_tree stochastic completions per tree for a total of T * m_per_tree completed datasets. Each completed dataset is conditional on a specific posterior tree (recorded in mi$tree_index).

Usage

```
multi_impute_trees(
  traits,
  trees,
  m_per_tree = 1L,
  species_col = NULL,
  trait_types = NULL,
  multi_proportion_groups = NULL,
  log_transform = TRUE,
  missing_frac = 0.25,
  covariates = NULL,
  epochs = 2000L,
  verbose = TRUE,
  seed = NULL,
  share_gnn = TRUE,
  reference_tree = NULL,
  draws_method = c("mc_dropout", "conformal"),
  gnn = FALSE,
  ...
)
```

Arguments

traits	data.frame. Same format as multi_impute() and impute().
trees	list of phylo objects (class multiPhylo or plain list). Each tree must contain the species in traits as tips. Posterior samples from BirdTree.org (Jetz et al. 2012) are ideal; the bundled trees300 dataset provides 50 posterior trees for avonet300.
m_per_tree	integer. Number of stochastic completion draws per tree (default 1). Total datasets = length(trees) * m_per_tree.
species_col	character or NULL. See impute().
trait_types	named character vector overriding auto-detected trait types. Required for "proportion" and "zi_count". See impute() / preprocess_traits(). Default NULL (auto-detect).
multi_proportion_groups	named list declaring compositional trait groups (rows summing to 1), forwarded to impute() / preprocess_traits(). Default NULL.
log_transform	logical. Log-transform every continuous trait whose observed values are all positive (default TRUE), then back-transform the imputations. Set it to FALSE when traits are already on a log scale (for example log body mass): otherwise they are logged a second time and imputed on a log(log) scale, which can bias analyses done on the log scale. With a mix of raw and already-logged traits, log the raw ones yourself and set FALSE. In a real-data check (PanTHERIA, 4,027 mammal species, four already-logged traits, posterior draws), the default shifted Rubin-pooled regression slopes by up to 3.9 standard errors of the complete-data slope, in a direction that depended on which values were missing; with FALSE the largest shift was 1.8.
missing_frac	numeric. Fraction held out for validation/test during training (default 0.25).
covariates	data.frame or matrix of environmental covariates (fully observed, numeric). Passed through to impute(). Default NULL (no covariates).
epochs	integer. Maximum GNN training epochs per tree (default 2000).

verbose	logical. Print progress (default TRUE).
seed	optional integer. Base random seed; when supplied, each tree uses seed + t - 1 for reproducible results. The default NULL uses the current RNG stream.
share_gnn	logical. With gnn = TRUE, TRUE (the default) trains one GNN on a reference tree and reuses it while recomputing the baseline for each posterior tree. Set FALSE to refit the GNN on each tree. With the default gnn = FALSE, no GNN is trained in either mode. See the "Share-GNN" section below.
reference_tree	optional phylo used for the reference fit when share_gnn = TRUE. Default NULL selects the maximum-clade-credibility tree via phangorn::maxCladeCred(trees). If phangorn is not installed, falls back to trees[[1]] with a warning.
draws_method	character. How the m_per_tree stochastic draws are generated for each tree (P1-11). One of "mc_dropout" (default, back compatible with pre-P1-11 behaviour) or "conformal". See the "Which draw mechanism this uses" section below for the trade-off between the two.
gnn	logical. Passed through to impute() / fit_pigauto() for every tree. When TRUE, the usual GNN correction is trained (once, on the reference tree, when share_gnn = TRUE). When FALSE, no GNN is used anywhere in the pipeline: baseline-only fits throughout (the default since 0.11.0.9001; see impute()). With share_gnn = TRUE, each tree's baseline is fit without held-out splits (splits = NULL), matching the production baseline_full semantics fit_pigauto() uses.
...	additional arguments forwarded to fit_pigauto() via impute().

Details

This is an experimental prediction-sensitivity workflow. It compares stochastic completions across a posterior tree sample; it does not provide a validated downstream inferential workflow. See vignette("tree-uncertainty").

Every completed dataset carries a different tree's signal so that between-tree variation enters the returned stochastic completions. This path is not supported by multi_impute_analysis() and has not passed the analysis-aware inferential gate. Do not use it for downstream inference.

For each tree the function runs preprocessing, baseline fitting and prediction. A GNN is also fitted only when gnn = TRUE; with share_gnn = TRUE, that GNN is trained once on a reference tree. Topologies and branch lengths vary across trees, so the phylogenetic baseline covariance differs for each tree.

The returned datasets may be compared descriptively to assess sensitivity of point imputations to the tree sample. No calibrated downstream standard error or Rubin-pooling claim is made for this path.

Computation time. The following historical planning estimates apply to gnn = TRUE, with one shared GNN fit plus T baseline passes when share_gnn = TRUE. They are not timings for the current GNN-off default; estimate the intended tree and sample setting before a campaign.

Species n	1 fit	T = 50 share_gnn=TRUE	T = 50 share_gnn=FALSE
300	~30-60 s	~3-5 min	25-50 min
5,000	~5-10 min	~10-20 min	4-8 hr
10,000	~20-40 min	~30-60 min	17-33 hr

Value

An object of class "pigauto_mi_trees", inheriting from "pigauto_mi", with components:

`datasets` List of `T * m_per_tree` completed data.frames. Observed cells are preserved; missing cells are filled with stochastic draws for prediction-sensitivity diagnostics.

`m` Total number of datasets (`T * m_per_tree`).

`n_trees` Number of posterior trees used.

`m_per_tree` Imputations per tree.

`tree_index` Integer vector of length `m`; element `i` gives the tree index (1..T) for dataset `i`.

`pooled_point` Single data.frame averaging across all `T * m_per_tree` datasets. For reporting, not inference.

`se` Matrix of per-cell pooled SEs (NA if not available).

`imputed_mask` Logical matrix; `TRUE` where a cell was originally missing.

`share_gnn` Logical; `TRUE` if the shared-GNN path was used.

`draws_method` Character; `"mc_dropout"` or `"conformal"`, echoing the argument this call used (P1-11).

`mi_workflow` `"pigauto_tree_sensitivity_diagnostic"`, recording that these are prediction-sensitivity completions and cannot be passed to `with_imputations()` or `pool_mi()`.

`fit` Single `pigauto_fit` fitted on the reference tree when `share_gnn = TRUE`; `NULL` otherwise.

`fits` List of `T` `pigauto_fit` objects (one per tree) when `share_gnn = FALSE`; `NULL` when `share_gnn = TRUE`.

`reference_tree` The reference `phylo` used for the shared fit when `share_gnn = TRUE`; `NULL` otherwise.

`trees` The input posterior trees.

`species_col` Passed-through species column name.

When to use this

Choose the prediction-diagnostic function based on how many trees you have:

- **One tree** (single published phylogeny, single time-calibrated tree): use `multi_impute()` and select a draw method explicitly when needed.
- **Multiple posterior trees** (BirdTree samples, BEAST posterior, etc.): use `multi_impute_trees()` only for prediction sensitivity. Tree uncertainty is unsupported by the analysis-aware backend.

Share-GNN (tree-sharing) mode

When `gnn = TRUE, share_gnn = TRUE`, the GNN weights and spectral features are trained once on the reference tree (MCC by default). For each posterior tree the BM / joint-MVN baseline is recomputed, and the prediction is the blend `(1 - r_cal) * baseline_t + r_cal * gnn_shared`. Because `r_cal` is calibrated once on held-out data at the reference tree and applied uniformly, the tree-uncertainty contribution is:

- Fully preserved when the gate is closed (r_cal near 0): the GNN contributes nothing, and the baseline varies per tree.
- Partially preserved when the gate is open: the baseline portion still varies, but the GNN portion is a tree-invariant constant — this slightly under-estimates tree variance in the GNN channel.
- Lost in the GNN channel when the gate is fully open (rare on real data; the baseline channel still carries tree variation).

On every real dataset benchmarked in the v0.9.0 campaign the gate closed partially or fully. This evidence is specific to those benchmark regimes and does not guarantee tree-variance calibration elsewhere. Set `share_gnn = FALSE` with `gnn = TRUE` if you need a separate GNN fit for each tree. With the default `gnn = FALSE`, there is no GNN channel.

Which draw mechanism this uses (and its limitation)

The default `draws_method = "mc_dropout"` passes `n_imputations = m_per_tree` to `impute()`. With `gnn = TRUE`, each draw combines a fresh baseline draw with a stochastic GNN pass. With the default `gnn = FALSE`, no GNN or dropout runs; only the baseline draw varies within each tree.

This differs from the conformal option in `multi_impute()`. That function now defaults to `draws_method = "auto"`, selecting posterior draws for eligible continuous-trait inputs and conformal draws otherwise. Baseline and optional GNN draws reflect model and prior uncertainty, whereas conformal draws use held-out prediction residuals. Neither mechanism validates downstream pooling or inference for posterior-tree completions.

`multi_impute_trees()` now exposes a `draws_method` argument (P1-11). `draws_method = "conformal"` runs `impute` with `n_imputations = 1` on each tree and draws the `m_per_tree` completions from the calibrated conformal scores instead, via the same internal sampling helper `multi_impute()` uses. Default remains `"mc_dropout"` for back compatibility. Within-tree calibration changes the within-tree stochastic completion mechanism; both settings are prediction-sensitivity diagnostics. Between-tree variation is present under either setting because the baseline (and, when `share_gnn = FALSE`, the GNN) still varies per tree.

Safety floor + share_gnn (v0.9.1.9002+)

When `share_gnn = TRUE` with `safety_floor = TRUE`, the grand-mean baseline `mean_baseline_per_col` and the three calibrated weights (`r_cal_bm`, `r_cal_gnn`, `r_cal_mean`) are computed ONCE on the reference tree and reused across all posterior trees. They are properties of the observed training traits, not of the tree topology. Each posterior tree only recomputes the BM baseline; the GNN delta and the three weights stay fixed. This preserves prediction sensitivity to the per-tree baseline without re-calibrating the safety floor; it does not establish downstream inferential validity.

References

Nakagawa S, de Villemereuil P (2019). "A general method for simultaneously accounting for phylogenetic and species sampling uncertainty via Rubin's rules in comparative analysis." *Systematic Biology* 68(4): 632-641.

Jetz W, Thomas GH, Joy JB, Hartmann K, Mooers AO (2012). "The global diversity of birds in space and time." *Nature* 491(7424): 444-448.

See Also

`multi_impute()` for single-tree prediction diagnostics or, with `draws_method = "posterior"`, single-tree inference on continuous traits; `multi_impute_analysis()` for analysis-aware MI; and trees300

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
trees <- structure(list(tree, tree), class = "multiPhylo")
df <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                c("Mass", "Wing.Length"), drop = FALSE]
rownames(df) <- tree$tip.label
df$Mass[seq_len(3L)] <- NA_real_
mi <- multi_impute_trees(df, trees, m_per_tree = 1L, epochs = 5L,
                         verbose = FALSE)
print(mi)
mass_by_tree <- vapply(mi$datasets, function(dat) dat$Mass,
```

```
                        numeric(nrow(df)))
apply(mass_by_tree, 1L, stats::sd)
```

`pigauto_report` *Generate an HTML imputation report from a pigauto fit*

Description

Produces an HTML file with interactive charts (loading Chart.js from its public CDN). For fits created with the default `gnn = FALSE`, the report describes baseline-only imputation. For fits created with `gnn = TRUE`, it also compares the GNN against the phylogenetic baseline and includes training history. Per-trait metrics, gate values, and held-out diagnostic conformal coverage are included when available.

Usage

```
pigauto_report(
  fit,
  data = NULL,
  splits = NULL,
  output_path = "pigauto_report.html",
  title = "pigauto Imputation Report",
  open = TRUE
)
```

Arguments

`fit`	A `pigauto_fit` object (or a `pigauto_result` from `impute`).
`data`	Optional `pigauto_data` object. Extracted automatically when `fit` is a `pigauto_result`. When supplied it must match the fit (species and traits).
`splits`	Optional splits object from `make_missing_splits`. Extracted automatically when `fit` is a `pigauto_result`. When supplied it must match the fit.
`output_path`	Character of length 1. File path for the HTML report (default `"pigauto_report.html"` in the working directory).
`title`	Character of length 1. Report title (HTML-escaped on write).
`open`	Logical of length 1. Open the report in a browser when done (default `TRUE`).

Value

The output path (invisibly).

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
traits$Mass[seq_len(3L)] <- NA_real_
```

```
result <- impute(traits, tree, epochs = 5L, verbose = FALSE)
pigauto_report(result, output_path = tempfile(fileext = ".html"),
               open = FALSE)
```

`plot.pigauto_benchmark` *Plot a pigauto benchmark*

Description

Creates a multi-panel comparison plot showing BM baseline vs pigauto performance across all simulation scenarios.

Usage

```
## S3 method for class 'pigauto_benchmark'
plot(x, metric = "rmse", ...)
```

Arguments

`x`	pigauto_benchmark object.
`metric`	character. Which metric to plot (default `"rmse"`).
`...`	passed to base plot functions.

Value

Invisible `NULL`.

Examples

```
benchmark <- structure(list(summary = data.frame(
  scenario = rep("BM", 2), trait = rep("trait1", 2),
  method = c("baseline", "pigauto"), metric = rep("rmse", 2),
  mean = c(1, 0.8)
)), class = "pigauto_benchmark")
plot(benchmark)
```

`plot.pigauto_fit` *Plot diagnostics for a fitted pigauto model*

Description

S3 `plot` method for `pigauto_fit` objects, using base R graphics. Three plot types are available: training history, calibrated gate values, and conformal prediction scores.

Usage

```
## S3 method for class 'pigauto_fit'
plot(x, type = "history", ...)
```

Arguments

x	An object of class "pigauto_fit".
type	Character. "history" (default when a GNN was trained): 2x2 panel of training loss components (reconstruction, shrinkage, gate regularisation) and validation loss over epochs. A fit with gnn = FALSE (the default) has no training history, so plot(fit) then shows "gates" instead. "gates": bar plot of calibrated gate values per trait, coloured by trait type (green = continuous, blue = count, orange = ordinal, red = binary, purple = categorical). "conformal": bar plot of conformal prediction scores per trait with a reference line at the median score.
...	Additional arguments passed to base plotting functions.

Value

Invisible NULL. Called for its side effect (plotting).

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
data <- preprocess_traits(traits, tree)
splits <- make_missing_splits(data$X_scaled, trait_map = data$trait_map)
# The package default disables the optional GNN, so this example works
# without torch or its Lantern runtime.
fit <- fit_pigauto(data, tree, splits = splits, epochs = 5L,
                   eval_every = 1L, verbose = FALSE, gnn = FALSE)
plot(fit)
plot(fit, type = "gates")
plot(fit, type = "conformal")
```

---

plot.pigauto_pred *Plot predictions from a pigauto model*

Description

S3 plot method for pigauto_pred objects, using base R graphics. Three plot types are available: scatter plots of observed vs predicted values, interval plots per species, and probability charts for discrete traits.

Usage

```
## S3 method for class 'pigauto_pred'
plot(x, data = NULL, splits = NULL, type = "scatter", trait = NULL, ...)
```

Arguments

x	An object of class "pigauto_pred".
data	Optional numeric matrix or data.frame of observed values in original scale. When supplied, observed values are overlaid on scatter and interval plots.
splits	Optional list (output of make_missing_splits). Used to identify which cells were held out for evaluation.
type	Character. "scatter" (default): grid of observed-vs-predicted scatter plots for continuous, count, and proportion traits, with 1:1 line and conformal interval bands where available. "intervals": for each trait, species sorted by predicted value with conformal-interval ribbon; observed values shown in red where available. "probabilities": for binary traits, boxplot of predicted probabilities grouped by true class; for categorical traits, stacked bar chart of mean predicted probabilities by true class.
trait	Character vector of trait names to plot. If NULL (default), all traits appropriate for the chosen type are plotted.
...	Additional arguments passed to base plotting functions.

Value

Invisible NULL. Called for its side effect (plotting).

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
traits$Mass[seq_len(3L)] <- NA_real_
pred <- impute(traits, tree, epochs = 5L, verbose = FALSE)$prediction
plot(pred)
plot(pred, type = "scatter", data = traits)
plot(pred, type = "intervals", trait = "Mass")
```

---

plot.pigauto_result *Plot diagnostics for a pigauto result*

Description

S3 plot method for pigauto_result objects. Delegates to plot.pigauto_fit on x$fit.

Usage

```
## S3 method for class 'pigauto_result'
plot(x, ...)
```

Arguments

x	An object of class "pigauto_result".
...	Passed to plot.pigauto_fit.

Value

Invisible NULL. Called for its side effect (plotting).

`plot_comparison` *Forest-plot style comparison of benchmark results*

Description

Takes a data.frame of benchmark results and produces a forest-plot style figure comparing baseline and GNN performance per trait. Each trait appears on the y-axis; points mark RMSE (continuous/count) or accuracy (binary/categorical) for the two methods, connected by a horizontal line colour-coded by whether the GNN improves over the baseline.

Usage

```
plot_comparison(
  results,
  metric = NULL,
  methods = c("baseline", "pigauto"),
  ...
)
```

Arguments

`results`	A data.frame with columns `trait`, `type`, `metric`, `method`, and `value`. Typically the output of `compare_methods`, `evaluate`, or reshaped output from `evaluate_imputation`.
`metric`	Character. Which metric to compare. Default `NULL` auto-selects: `"rmse"` for continuous/count traits and `"accuracy"` for binary/categorical traits. If a single metric name is supplied, only traits with that metric are shown.
`methods`	Character vector of length 2: the baseline method name and the pigauto method name. Default `c("baseline", "pigauto")`, matching the labels returned by `compare_methods` and `evaluate`.
`...`	Additional arguments passed to `plot()`.

Details

When both RMSE and accuracy metrics are present and `metric` is `NULL`, a two-panel figure is produced automatically.

Value

Invisible NULL. Called for its side effect (plotting).

Examples

```
results <- data.frame(
  trait = rep("Mass", 2L), type = rep("continuous", 2L),
  metric = rep("rmse", 2L), method = c("baseline", "pigauto"),
  value = c(1, 0.8)
)
plot_comparison(results)
plot_comparison(results, metric = "rmse")
```

plot_history_gg *Plot training history (ggplot2, deprecated)*

Description

Deprecated. Use the base-R S3 method instead: `plot(fit, type = "history")`.

Produces a ggplot2 validation loss training curve.

Usage

```
plot_history_gg(x, ...)
```

Arguments

`x`	object of class `"pigauto_fit"`.
`...`	ignored.

Value

A `ggplot` object.

Examples

```
fit <- structure(
  list(history = data.frame(epoch = 1:3, val_loss = c(1, 0.7, 0.5))),
  class = "pigauto_fit"
)
plot_history_gg(fit)
```

plot_uncertainty *Plot uncertainty ribbons for imputed trait values*

Description

Plots conformal prediction intervals when present, otherwise an approximate `prediction +/- 1.96 * SE` ribbon, sorted by predicted value. If `truth` is supplied, observed values are overlaid as points. Works for continuous, count, and ordinal traits. For binary traits, plots predicted probabilities with an uncertainty ribbon.

Usage

```
plot_uncertainty(pred_result, truth = NULL, trait_name)
```

Arguments

`pred_result`	list with `imputed` and `se` components (output of `predict.pigauto_fit`), or a `pigauto_result` from `impute` (its `$prediction` is used).
`truth`	matrix or data.frame of true values (same scale as `pred_result$imputed`), or `NULL`.
`trait_name`	character. Which trait to plot (must match column name).

Value

A ggplot object.

Examples

```
pred <- list(
  imputed = data.frame(mass = c(10, 12, 15)),
  se = matrix(c(1, 1.5, 2), ncol = 1,
              dimnames = list(NULL, "mass"))
)
plot_uncertainty(pred, trait_name = "mass")
```

`pool_mi` *Pool downstream model fits across multiple imputations (Rubin's rules)*

Description

Combine regression coefficients from `M` model fits – one per imputed dataset – into a single pooled table using Rubin's rules. The pooled standard errors combine within-imputation sampling variance and between-imputation variance. Correct Rubin arithmetic does not make an incompatible imputation model inferentially valid. For pigauto inference, fits should come from the documented `multi_impute_analysis()` workflow, or, for continuous traits, from `multi_impute()` with `draws_method = "posterior"` (see `with_imputations()` for the analyses those draws cover), via `with_imputations()`. The backend is experimental and supports fixed-effect coefficients only. Variance components, correlations, random-effect predictions, BLUPs/conditional modes, latent loadings, and other structured parameters are unsupported.

Usage

```
pool_mi(
  fits,
  conf.level = 0.95,
  coef_fun = NULL,
  vcov_fun = NULL,
  df_fun = NULL,
  tidy_fun = NULL
)
```

Arguments

`fits`	A list of model fits of length `M >= 2`. Automatic fixed-effect adapters are provided for `stats::lm`, `stats::glm`, `nlme::gls`, `nlme::lme`, `lme4::merMod`, `glmmTMB::glmmTMB`, `drmTMB`, and `gllvmTMB_multi` fits. Other classes implementing `coef()` and `vcov()` also work. The output of `with_imputations()` is accepted directly. Bayesian fits (`brms`, `MCMCglmm`) are rejected; see Details. Bare user-supplied fit lists are accepted with an explicit warning because their imputation provenance cannot be verified. pigauto workflow classes and markers are caller-controlled metadata; they do not authenticate the fits or their history.
`conf.level`	Confidence level for the pooled interval (default `0.95`).

`coef_fun` Optional function extracting a named numeric fixed-effect vector from one fit. `NULL` uses the automatic class adapter. Custom coefficient and covariance extractors can be supplied independently; callers must ensure custom extractors select fixed effects only. For recognized `drmTMB` and `gllvmTMB_multi` fits, pigauto still requires convergence code zero and a positive-definite Hessian, even when a custom extractor is supplied.

`vcov_fun` Optional function extracting the fixed-effect covariance matrix. `NULL` uses the automatic class adapter. Base matrices and `Matrix` objects are accepted; the selected covariance block must be square, finite, symmetric, and have non-negative diagonal entries.

`df_fun` Optional function returning the complete-data residual degrees of freedom `nu_com` from one fit. When supplied, pooled degrees of freedom use the Barnard & Rubin (1999) small-sample correction, which is less biased for short series. When `NULL` (the default) the classical Rubin (1987) formula is used.

`tidy_fun` Optional function returning a data.frame with unique `term`, numeric `estimate`, and non-negative finite `std.error` columns. This is an alternative to `coef_fun` and `vcov_fun`, not a supplement; combining them is an error. Callers must ensure it selects fixed effects only.

Details

Let $\hat{\theta}_i$ be the coefficient vector from fit i and $U_i = \mathrm{vcov}(\mathrm{fit}_i)$, for $i = 1, \ldots, M$. Rubin's rules (Rubin 1987) give

$$\bar{\theta} = M^{-1} \sum_i \hat{\theta}_i$$

$$W = M^{-1} \sum_i \mathrm{diag}(U_i)$$

$$B = (M-1)^{-1} \sum_i (\hat{\theta}_i - \bar{\theta})^2$$

$$T = W + (1 + 1/M)B$$

with pooled standard error $\sqrt{T}$. The relative increase in variance is $r = (1 + 1/M)B/W$, the classical pooled df is $\nu_{\mathrm{old}} = (M-1)(1 + 1/r)^2$, and the fraction of missing information is

$$\mathrm{fmi} = (r + 2/(\nu + 3))/(r + 1).$$

When `df_fun` returns finite complete-data df `nu_com`, the Barnard-Rubin (1999) correction combines $\nu_{\mathrm{obs}} = ((\nu_{\mathrm{com}} + 1)/(\nu_{\mathrm{com}} + 3))\nu_{\mathrm{com}}(1 - \lambda)$ with `nu_old` via $\nu_{\mathrm{BR}} = 1/(1/\nu_{\mathrm{old}} + 1/\nu_{\mathrm{obs}})$. With no between-imputation variation (`B = 0`), the classical limit is `df = Inf`, `riv = 0`, and `fmi = 0`. If finite complete-data df are supplied, Barnard–Rubin instead retains finite observed-data df and its small-sample FMI adjustment. A completely deterministic quantity (`B = W = 0`) always has zero FMI and infinite df.

Only fixed-effect coefficients are pooled. Random-effect variances and correlations, BLUPs/conditional modes, latent loadings, and other structured parameters require parameter-specific transformations and are not supported by the automatic `pool_mi()` adapters. Custom extractors are an expert escape hatch and cannot be inspected by pigauto; using them to select unsupported structured parameters is outside the documented scope. The `glmmTMB` adapter selects conditional fixed effects only. The `drmTMB` adapter includes named distributional fixed-effect blocks such as regression coefficients for `mu` and `sigma`; those are fixed coefficients, not random-effect variance components. `drmTMB` and `gllvmTMB` need to be installed and loadable only when their automatic adapters are used. For either recognized backend class, pooling also requires the fit to report scalar convergence code `0` and

pdHess = TRUE; missing or invalid status is rejected before coefficient extraction, including with custom extractors.

Supported classes. lm, glm, gls, lme, merMod (lme4), glmmTMB (conditional component), drmTMB, and gllvmTMB_multi are handled automatically. Other classes are pooled through their coef() / vcov() methods, or through coef_fun / vcov_fun / tidy_fun.

Bayesian fits (brms, MCMCglmm) are rejected because Rubin's rules are not the right tool for posterior samples: they are pooled by concatenating posterior draws across imputations. For brms, use brms::brm_multiple(formula, data = mi$datasets), or brms::combine_models() on a list of existing fits. For MCMCglmm, concatenate the Sol samples across fits. No Bayesian downstream workflow is supported by the analysis-aware backend.

Value

A data.frame with one row per coefficient and columns:

term Coefficient name.

estimate Pooled point estimate (mean across fits).

std.error Pooled standard error sqrt(T) where T = W + (1 + 1/M) * B.

df Pooled degrees of freedom (Barnard-Rubin if df_fun supplied, else classical Rubin).

statistic estimate / std.error.

p.value Two-sided p-value from a t distribution on df.

conf.low, conf.high Pooled conf.level interval.

fmi Fraction of missing information.

riv Relative increase in variance due to non-response.

References

Rubin DB (1987). *Multiple Imputation for Nonresponse in Surveys.* Wiley.

Barnard J, Rubin DB (1999). "Small-sample degrees of freedom with multiple imputation." *Biometrika* 86(4): 948-955.

Nakagawa S, Freckleton RP (2008). "Missing inaction: the dangers of ignoring missing data." *Trends in Ecology & Evolution* 23(11): 592-596.

Nakagawa S, Freckleton RP (2011). "Model averaging, missing data and multiple imputation: a case study for behavioural ecology." *Behavioral Ecology and Sociobiology* 65(1): 103-116.

See Also

multi_impute_analysis(), with_imputations()

Examples

```
dat <- data.frame(y = stats::rnorm(30L), x = stats::rnorm(30L),
                  z = stats::rnorm(30L))
dat$x[seq(3L, 30L, by = 5L)] <- NA_real_
mi <- multi_impute_analysis(
  data = dat, formula = y ~ x + z, missing = "x",
  model = "lm", m = 2L
)
fits <- with_imputations(mi, function(d) lm(y ~ x + z, data = d))
pool_mi(fits)
```

`predict.pigauto_fit` *Impute missing traits using a fitted pigauto model*

Description

Runs a single forward pass through the fitted model and returns imputed trait values back-transformed to the original scale. Supports all trait types (continuous, binary, categorical, ordinal, count, proportion, zero-inflated count, multi-proportion) and MC dropout for stochastic prediction diagnostics (when `n_imputations > 1`). The fitted model is a gated ensemble of a phylogenetic baseline and a graph neural network correction; prediction is the per-trait blend `(1 - r_cal) * baseline + r_cal * delta_GNN`.

Usage

```
## S3 method for class 'pigauto_fit'
predict(
  object,
  newdata = NULL,
  return_se = TRUE,
  n_imputations = 1L,
  baseline_override = NULL,
  pool_method = c("median", "mean", "mode"),
  clamp_outliers = FALSE,
  clamp_factor = 5,
  match_observed = c("none", "pmm"),
  pmm_K = 5L,
  ...
)
```

Arguments

`object`	object of class `"pigauto_fit"`.
`newdata`	`NULL` (use the training data) or a `"pigauto_data"` object for new species.
`return_se`	logical. Compute standard errors? (default `TRUE`).
`n_imputations`	integer. Number of stochastic imputation draws — BM posterior samples plus GNN dropout — (default `1L`). Set to e.g. 10 or 20 to inspect stochastic prediction variation. These are not validated analysis-aware multiple imputations.
`baseline_override`	optional `list(mu, se)` with the same shape as `object$baseline`. When supplied, predictions use this baseline instead of the one saved in the fit. Used internally by `multi_impute_trees()` to reuse a trained GNN across posterior trees. Most users can ignore this. Default `NULL` (use the fit's own baseline).
`pool_method`	character. How to pool the M decoded draws when `n_imputations > 1`: `"median"` (default) takes the per-cell median for count / proportion / zi_count magnitude traits — robust to single dropout-noisy draws amplified by non-linear decoders (`expm1` / `plogis`). `"mean"` restores the pre-v0.9.2 arithmetic-mean pooling. Continuous / ordinal / binary / categorical pooling is unchanged (linear / probability-averaged). See issue #40.

`clamp_outliers` logical. Phase G (v0.9.1.9011+). When `TRUE`, post-back-transform predictions for log-transformed continuous, count, and zi_count magnitude traits are capped at `tm$obs_max * clamp_factor` (`tm$obs_max` is the observed maximum on the original scale, set at preprocess time). Targets tail-extrapolation modes amplified by `exp()` / `expm1()` back-transforms. Default `FALSE` preserves v0.9.1 behaviour exactly.

`clamp_factor` numeric scalar (>= 1). Multiplicative factor on the observed maximum used by `clamp_outliers`. Default 5. Ignored when `clamp_outliers = FALSE`.

`match_observed` character, one of `c("none", "pmm")`. Phase G' (v0.9.1.9012+). When `"pmm"`, uses Predictive Mean Matching (Little 1988; Buuren mice) on the at-risk types (log-transformed continuous, count, zi_count magnitude, proportion). For each missing cell, finds the `pmm_K` observed cells whose own predictions are closest to the missing cell's prediction, samples one, and returns its observed value. Imputed values are guaranteed to lie in the observed data range – no extrapolation is possible by construction.

When to use: PMM is a niche feature in pigauto. The package already provides conformal prediction intervals (calibrated against held-out residuals) and stochastic conformal-width and Brownian/MC-dropout prediction draws. PMM is only worth enabling for: (a) methodological comparison against mice / equivalent packages, or (b) workflows that specifically require imputed values to come from the observed data pool. For tail safety on single-imputation point estimates, prefer `clamp_outliers = TRUE`. PMM failed the downstream fixed-effect redesign pilot and is unsupported for inference.

The Phase G' acceptance bench (`useful/MEMO_2026-05-01_phase_g_prime_results.md`) confirmed PMM does not strictly improve point-estimate RMSE over the no-PMM default: it wins on extrapolating cells (e.g. AVONET Casuarius) but loses on cells where the GNN's prediction is already accurate (donor-mismatch noise).

Discrete-class types (binary / categorical / ordinal / multi_proportion) and un-log continuous: no-op. Default `"none"` preserves v0.9.1.9011 behaviour exactly.

`pmm_K` integer scalar (>= 1). Donor pool size for PMM. Default `5L` (mice convention). Ignored when `match_observed = "none"`.

`...` ignored.

Details

When `n_imputations > 1`, each imputation `m` draws a BM posterior sample `t_BM_draw ~ N(BM_mu, BM_se)` on the latent scale for originally-missing cells (`BM_se = 0` for observed cells so they are never perturbed). The model runs in train mode (GNN dropout active) using `t_BM_draw` as input. The final blend is `(1 - r_cal) * t_BM_draw + r_cal * GNN_delta(t_BM_draw)`: when the calibrated gate is zero the imputation is a pure BM posterior draw; when `r_cal > 0` both BM draws and GNN dropout contribute variance. Point estimates are the mean (continuous, count) or mode (binary, categorical, ordinal) across passes. The M complete datasets are returned in `imputed_datasets` for prediction diagnostics. These datasets are not supported for downstream inference. Use `multi_impute_analysis()` for the narrow analysis-aware backend after consulting its supported-model and lifecycle documentation.

`gnn = FALSE` **fits (traditional-stats mode):** when the fit was produced with `gnn = FALSE`, no GNN was ever trained, so this function makes no `torch::` calls at all – prediction is the calibrated baseline blend `r_cal_bm * baseline + r_cal_mean * mean_baseline_per_col` (`r_cal_gnn` is always zero) computed on plain R matrices. Two baselines are available: in production mode (the default, no `.mask_observed_idx`) predictions use `baseline_full` – fit with `splits = NULL`, so every observed cell can inform every other cell; in evaluation mode (`.mask_observed_idx` supplied, as used

internally for val/test scoring) predictions use the held-out `baseline` instead, so scored cells never see their own value. `n_imputations > 1` draws are BM posterior samples `MU_m ~ N(baseline_mu, baseline_se)` at originally-missing cells (observed cells are always restored to their true value), with no GNN dropout contribution.

Decoding per type:

continuous reverse z-score, then `exp()` if log-transformed

binary `sigmoid(latent)` to probability, round to 0/1

count reverse z-score of log1p, `expm1()`, round, clip >= 0

ordinal reverse z-score, round to nearest valid integer level

categorical `softmax()` over K latent columns, argmax

Conformal intervals: `conformal_lower` / `conformal_upper` use whatever `conformal_method` the fit was trained with (`fit_pigauto`). For `"mondrian"`, the half-width is not constant across rows: each target cell's phylogenetic locality (mean cophenetic distance to its 5 nearest training-observed species for that trait) picks the near- or far-stratum score stored on the fit, so predictions in undersampled clades get wider intervals. See `fit_pigauto`'s `conformal_method` argument for details.

`zi_count` **conformal intervals are conditional on non-zero.** `conformal_lower[, nm]` / `conformal_upper[, nm]` for a `zi_count` trait back-transform the magnitude latent (log1p-z of the count *given it is non-zero*) plus/minus the conformal half-width that was calibrated on that same magnitude column (`fit_pigauto`'s `compute_conformal_scores()` scores zi_count on `latent_cols[2]`, never on the decoded `E[X]`). They are **not** an interval for the expected value `E[X] = P(nonzero) * E[count | nonzero]` – constructing a naive ±quantile interval directly on `E[X]` would conflate gate and magnitude uncertainty into one number with no valid coverage interpretation. The gate probability `P(nonzero)` is available separately in `pred$probabilities[[nm]]`; combine it with the conditional interval yourself if you need an `E[X]`-scale bound.

Value

A list of class `"pigauto_pred"` with:

imputed data.frame of imputed values in original scale with proper R types (numeric, integer, factor, ordered).

imputed_latent Numeric matrix (n x p_latent) of predictions in latent scale.

se Numeric matrix (n x n_original_traits) of per-cell uncertainty. See the "What `$se` means per trait type" section below for what each trait type's column actually contains. `NULL` if `return_se = FALSE`.

probabilities Named list. Binary traits: numeric probability vector. Categorical traits: n x K probability matrix. Other types: not present.

imputed_datasets List of M stochastic prediction data.frames when `n_imputations > 1`; `NULL` otherwise. Not supported for downstream inference.

trait_map Trait map from the fitted model.

species_names Character vector.

trait_names Character vector.

n_imputations Integer, number of imputations performed.

What $se means per trait type

pred$se is a single n x n_original_traits matrix, but its per-column meaning differs by trait type — do not treat every column as a Gaussian standard error. This table is the definitive reference; other pigauto documentation (e.g. impute's return value) points here instead of repeating it.

Type	Value
continuous / count / ordinal / proportion	model-dependent BM/joint conditional SD, delta-method back-transformed to o
binary	min(p, 1-p)
categorical	1 - max(p_k)
zi_count	delta-method SE of E[X] = P(nonzero) * E[count | nonzero]
multi_proportion	NA

These families are **incompatible objects sharing one column name**: only the continuous-family entries are standard errors in the Gaussian sense, and **none** of the binary / categorical / zi_count entries may be used in Rubin's-rules arithmetic. The supported downstream multiple-imputation routes start with multi_impute_analysis or, for continuous traits, multi_impute with draws_method = "posterior", then use with_imputations and pool_mi.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
data <- preprocess_traits(traits, tree)
splits <- make_missing_splits(data$X_scaled, trait_map = data$trait_map)
fit <- fit_pigauto(data, tree, splits = splits, epochs = 5L,
                   verbose = FALSE)
pred <- predict(fit, return_se = TRUE)
pred$imputed
pred$se
pred$probabilities
pred2 <- predict(fit, n_imputations = 2L)
pred2$imputed_datasets
```

---

preprocess_traits *Preprocess trait data: align to tree, encode into latent space*

Description

Aligns species in the trait data frame to the tree, detects or accepts trait types (continuous, binary, categorical, ordinal, count, proportion, zi_count), and encodes each trait into a continuous latent matrix.

Usage

```
preprocess_traits(
  traits,
  tree,
  species_col = NULL,
  trait_types = NULL,
  multi_proportion_groups = NULL,
  log_cols = NULL,
  log_transform = TRUE,
  center = TRUE,
  scale = TRUE,
  covariates = NULL
)
```

Arguments

traits	data.frame with species as row names (one row per species), or with a species column identified by species_col (potentially multiple rows per species). Columns may be numeric, integer, factor, ordered, character, or logical.
tree	object of class "phylo".
species_col	character. Name of the column in traits that identifies species. When supplied, traits may have multiple rows per species. The column is removed from trait columns before encoding. Default NULL uses row names (one row per species).
trait_types	named character vector overriding auto-detection, e.g. c(Mass = "continuous", Diet = "categorical"). Valid types: "continuous", "binary", "categorical", "ordinal", "count", "proportion", "zi_count". Proportion and zi_count are override-only (not auto-detected). Unspecified traits are auto-detected. Note that "multi_proportion" is NOT set here — use multi_proportion_groups instead.
multi_proportion_groups	named list declaring compositional (multi-proportion) trait groups. Each element is a character vector of column names whose row-wise values sum to 1 (e.g. list(diet = c("plants", "insects", "fish"))). The group name becomes a single trait in the output, encoded via centred log-ratio (CLR) + per-component z-score. Group names must NOT match any column in traits. Default NULL.
log_cols	character vector of continuous trait names to log-transform. Default NULL means auto-detect (log if all observed values are positive). Set to character(0) to disable.
log_transform	logical. Legacy parameter: if TRUE and log_cols is NULL, log-transform all continuous traits with all-positive values. Overridden by log_cols when both are supplied. Traits already on a log scale should not be logged again: set log_transform = FALSE or leave them out of log_cols (see impute).
center	logical. Subtract column means for continuous/count/ordinal? Default TRUE.
scale	logical. Divide by column SDs for continuous/count/ordinal? Default TRUE.
covariates	data.frame or numeric matrix of environmental covariates. Covariates are conditioners: they inform imputation but are not themselves imputed, so they must be **fully observed** (no NAs — if a variable has missing values, put it in traits instead). Rows are aligned to the same species / observation order as X_scaled after traits are matched to the tree. Single-obs: rownames should match tree$tip.label

(they are reordered to tip order). Multi-obs: include the same `species_col` to match observation rows by species identity. Unnamed covariates are assumed to share the input-row order of `traits` and receive the same drop / reorder. Matching `nrow` alone is not enough — unmatched species names are an error.

Numeric / integer columns z-scored automatically.

Factor / ordered columns one-hot encoded (K binary columns per factor with K levels). Column names become `"var.level"`.

Character / logical columns coerced to factor, then one-hot.

Default `NULL` (no covariates).

Details

When each species has one observation (the default), output rows match `tree$tip.label` order. When `species_col` is supplied, multiple observations per species are supported: the output matrix has one row per observation, plus an `obs_to_species` mapping for the GNN (which operates at species level).

Automatic type detection (when `trait_types = NULL`) follows the R class of each column — no user input is required for most data:

R class	pigauto type
`numeric` (non-integer)	continuous
`integer`	count
`factor` with 2 levels	binary
`factor` (unordered) with >2 levels	categorical
`ordered` / `factor(..., ordered = TRUE)`	ordinal
`character`	converted to `factor`, then binary or categorical
`logical`	binary (FALSE = 0, TRUE = 1)

Two types require an explicit override because they cannot be distinguished from their R class alone:

`"proportion"` A `numeric` column bounded 0–1 looks identical to continuous. Declare it explicitly: `trait_types = c(SurvivalRate = "proportion")`. Encoded via `qlogis(clamp(x, 0.001, 0.999))`.

`"zi_count"` An `integer` column with excess zeros looks identical to count. Declare it explicitly: `trait_types = c(Parasites = "zi_count")`. Encoded as a binary zero/non-zero gate plus `log1p`-z magnitude.

Practical examples of type assignment:

- Body mass (`numeric`, all positive) → `continuous`, auto-log-transformed.
- Clutch size (`integer`) → `count`.
- Migratory (`factor` with levels "Yes"/"No") → `binary`.
- Diet (`factor` with >2 levels) → `categorical`.
- IUCN status (`ordered` factor, LC < NT < VU < EN < CR) → `ordinal`. If left as an unordered factor, it becomes `categorical` — both are valid depending on the question.
- Parasite load (`integer` with many zeros) → needs `trait_types = c(Parasites = "zi_count")`.
- Survival rate (`numeric`, values in 0 to 1) → needs `trait_types = c(Survival = "proportion")`.

Latent encoding per type:

continuous optional `log()`, then z-score (1 latent column)

binary 0/1 encoding (1 latent column)

count `log1p()`, then z-score (1 latent column)

ordinal integer coding (0 to K-1), then z-score (1 latent column)

categorical one-hot encoding (K latent columns)

proportion `qlogis(clamp(x, 0.001, 0.999))`, then z-score (1 latent column)

zi_count gate (0/1) + `log1p`-z of non-zeros (2 latent columns)

multi_proportion centred log-ratio (CLR) + per-component z-score (K latent columns per group). Rows must sum to 1. Declared via the `multi_proportion_groups` argument, not `trait_types`.

Value

A list of class `"pigauto_data"` with components:

X_scaled Numeric matrix (n_obs x p_latent), latent encoding. When `species_col` is `NULL`, n_obs = n_species.

X_raw Numeric matrix of continuous traits after optional log but before z-scoring (for backward compatibility).

X_original Original data.frame (aligned to tree, before encoding).

means Named numeric vector of column means used for z-scoring (continuous/count/ordinal traits only).

sds Named numeric vector of column SDs.

species_names Character vector of unique species matching `tree$tip.label` order (length = n_species).

obs_species Character vector of species labels per observation (length = n_obs). When multi-obs, can have duplicates. `NULL` when `species_col` is `NULL`.

obs_to_species Integer vector (length = n_obs) mapping each observation to its species index in `species_names`. `NULL` when `species_col` is `NULL`.

n_species Integer, number of unique species.

n_obs Integer, number of observations (= n_species when single-obs).

multi_obs Logical, `TRUE` when multiple observations per species are present.

trait_names Character vector of original trait names.

latent_names Character vector of latent column names.

trait_map List of trait descriptors (see Details).

p_latent Integer, total number of latent columns.

log_transform Logical, legacy field (TRUE if any continuous trait was log-transformed).

Examples

```
# Single-obs per species (backward compatible)
data(avonet300, tree300, package = "pigauto")
traits <- avonet300
rownames(traits) <- traits$Species_Key
traits$Species_Key <- NULL
pd <- preprocess_traits(traits, tree300)
dim(pd$X_scaled)   # 300 x p_latent
```

```
# Multi-obs per species (via species_col)
pd2 <- preprocess_traits(avonet300, tree300, species_col = "Species_Key")
pd2$n_obs      # number of observations
pd2$n_species  # number of unique species
```

`print.pigauto_check` *Print a pigauto input check*

Description

Print a pigauto input check

Usage

```
## S3 method for class 'pigauto_check'
print(x, ...)
```

Arguments

`x`	A `pigauto_check` object.
`...`	Unused.

Value

`x`, invisibly.

`print.summary_pigauto_result`
Print a pigauto result summary

Description

Print a pigauto result summary

Usage

```
## S3 method for class 'summary_pigauto_result'
print(x, ...)
```

Arguments

`x`	A `summary_pigauto_result` object.
`...`	Unused.

Value

`x`, invisibly.

`pull_gbif_centroids` *Fetch species range-centroid covariates from GBIF*

Description

Pulls occurrence records from the Global Biodiversity Information Facility (GBIF) for each species in `species`, aggregates them to a median lat / lon centroid, and returns a data.frame ready to be passed to `impute` via its `covariates` argument.

Usage

```
pull_gbif_centroids(
  species,
  cache_dir = NULL,
  occurrence_limit = 500L,
  sleep_ms = 100L,
  verbose = TRUE,
  refresh_cache = FALSE,
  store_points = FALSE
)
```

Arguments

`species`	character vector of species names.
`cache_dir`	directory to cache per-species RDS files. `NULL` (default) disables caching — not recommended for production use.
`occurrence_limit`	integer, maximum number of occurrences to fetch per species (default 500; paginated if > 300).
`sleep_ms`	integer, polite delay between API calls in milliseconds (default 100).
`verbose`	logical, print progress every 50 species.
`refresh_cache`	logical, force re-fetch even when cache exists.
`store_points`	logical. When `TRUE`, persists the raw filtered lat/lon occurrence points in each species' cache RDS under the `points` field. Used by `pull_worldclim_per_species` for per-occurrence bioclim extraction. Default `FALSE` preserves the pre-v0.9.1.9006 cache format.

Details

Caching is strongly recommended — GBIF rate-limits anonymous calls and the per-species fetch is expensive. With `cache_dir` set, each species gets one RDS file; subsequent calls skip the API.

For each species: resolve taxon via `name_backbone`, fetch up to `occurrence_limit` records via `occ_search` (paginated at 300 per GBIF call), filter out records with `hasGeospatialIssues = TRUE` and `basisOfRecord` in `c("FOSSIL_SPECIMEN", "LIVING_SPECIMEN")`, drop out-of-range coordinates, then take the median latitude and longitude as the species centroid.

Species with zero post-filter records receive `NA` centroids; their rows are still included in the returned data.frame so it aligns with the input species list.

Requires the optional `rgbif` package (in DESCRIPTION `Suggests`). If `rgbif` is not installed the function errors with an installation message.

Value

A data.frame with columns `species`, `centroid_lat`, `centroid_lon`, `n_occurrences`. Rownames are set to `species`. NA centroids are returned for species with no GBIF hits; the caller should decide how to handle them (typical: drop or impute).

See Also

`impute` (pass the return value as `covariates`).

Examples

```
sp <- c("Quercus alba", "Pinus taeda", "Acer saccharum")
cov <- pull_gbif_centroids(sp, cache_dir = "script/data-cache/gbif")
cov_num <- cov[, c("centroid_lat", "centroid_lon"), drop = FALSE]
```

---

pull_worldclim_per_species

Fetch per-species bioclim covariates from WorldClim v2.1

Description

Extends `pull_gbif_centroids` by extracting 19 WorldClim v2.1 bioclim variables at each species' GBIF occurrence points, aggregating per species (median + IQR), and returning a data.frame ready for `impute` via its `covariates` argument.

On first call, downloads the WorldClim 10-arc-minute raster stack (~130 MB compressed, ~500 MB unzipped) to `worldclim_cache_dir`. A sentinel file makes subsequent calls fully offline.

Per-species extracts are also cached (one RDS per species in `worldclim_cache_dir/extracts/`), so repeated calls for the same species list are instantaneous.

Since v0.9.1.9006 (2026-04-30) extraction operates on the raw per-occurrence point list when `pull_gbif_centroids` was called with `store_points = TRUE`. For legacy caches written by older `pull_gbif_centroids` (centroid-only) calls, extraction silently falls back to the species centroid.

Requires the optional `terra` package (in DESCRIPTION `Suggests`). If `terra` is not installed the function errors with an installation message.

Usage

```
pull_worldclim_per_species(
  species,
  gbif_cache_dir,
  worldclim_cache_dir,
  resolution = "10m",
  verbose = TRUE,
  refresh_cache = FALSE
)
```

Arguments

species	character vector of species binomials.
gbif_cache_dir	character path, GBIF per-species cache directory (as written by pull_gbif_centroids).
worldclim_cache_dir	character path, directory for WorldClim rasters + per-species extract cache. Created if absent.
resolution	one of "10m" (default), "5m", "2.5m".
verbose	logical. Print progress every 50 species.
refresh_cache	logical. Force re-extract even when per-species cache exists.

Value

A data.frame with length(species) rows and columns:

- species
- bio1_median, bio1_iqr, ..., bio19_median, bio19_iqr
- n_extracted (integer, number of valid occurrence points that contributed)

Rownames are set to species. Species with no GBIF hits get all-NA bio values and n_extracted = 0.

References

Fick SE, Hijmans RJ (2017). WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. International Journal of Climatology 37, 4302–4315.

See Also

pull_gbif_centroids (B.1, required input), impute (consume as covariates).

Examples

```
sp <- c("Quercus alba", "Pinus taeda", "Acer saccharum")
gbif_df <- pull_gbif_centroids(sp,
  cache_dir = "script/data-cache/gbif")
wc_df   <- pull_worldclim_per_species(
  species             = sp,
  gbif_cache_dir      = "script/data-cache/gbif",
  worldclim_cache_dir = "script/data-cache/worldclim")
# Combine for impute()
cov <- cbind(gbif_df[, c("centroid_lat", "centroid_lon")],
             wc_df[, grep("^bio", colnames(wc_df))])
```

read_traits *Read trait data from a CSV file or data frame*

Description

Loads trait data and sets species names as row names. If a CSV path is supplied, it is read with `read.csv`. All species identifiers must be present and unique because row names identify one trait row per species. For repeated observations, use `read.csv()` and pass the resulting data frame with `species_col` to `check_pigauto()` and `impute()`. By default, all non-species columns are returned (numeric, factor, integer, etc.). Use `trait_cols` to select a subset.

Usage

```
read_traits(
  x,
  species_col = "species",
  trait_cols = NULL,
  factor_cols = NULL,
  ordered_cols = NULL
)
```

Arguments

`x`	character path to a CSV file, or a `data.frame`.
`species_col`	character. Name of the column containing species names (default `"species"`).
`trait_cols`	character vector of column names to include. If `NULL` (default), all columns except `species_col` are used.
`factor_cols`	character vector of columns to coerce to `factor`.
`ordered_cols`	character vector of columns to coerce to `ordered`.

Value

A `data.frame` with unique species names as row names.

Examples

```
df <- data.frame(species = c("Sp_a", "Sp_b"), mass = c(10, 20))
traits <- read_traits(df, species_col = "species")
```

read_tree *Read a phylogenetic tree from a file*

Description

Reads a Newick or NEXUS tree file using **ape**. Tries `ape::read.tree` first; falls back to `ape::read.nexus` if that fails.

Usage

```
read_tree(path)
```

Arguments

path	character. Path to the tree file.

Details

For bird phylogenies, BirdTree provides a species-subset download tool for up to 2,500 species and full-tree distributions for larger sets. Download and unzip the trees, then read a Newick tree file here. BirdTree documents its distributions as Newick trees; this function also accepts NEXUS files. A file with one tree returns a `"phylo"` object; a file with several trees returns a `"multiPhylo"` object.

pigauto does not download BirdTree data automatically. Pass a single-tree result to `impute(traits, tree)`. A multi-tree result can be passed to `multi_impute_trees(traits, trees)` for experimental prediction- sensitivity checks only; this path has not been validated for downstream inference.

BirdTree's subset tool defaults to at least 100 draws and recommends a reasonable sample (more than 100) for analyses using its full-tree distributions. pigauto's `multi_impute_trees()` remains experimental: it supports descriptive checks of prediction sensitivity only, and its stochastic datasets are not validated for downstream inference. See `vignette("tree-uncertainty")`.

BirdTree asks researchers using full or partial tree data to cite Jetz et al. (2012). If you use the BirdTree web tool, also credit BirdTree.org.

Value

An object of class `"phylo"` when the file contains one tree, or `"multiPhylo"` when it contains multiple trees.

References

Jetz W, Thomas GH, Joy JB, Hartmann K, Mooers AO (2012). The global diversity of birds in space and time. *Nature*, 491, 444-448. doi:10.1038/nature11631. BirdTree.org. Downloads, phylogeny subsets, and FAQ. https://birdtree.org/downloads/ https://birdtree.org/subsets/ https://birdtree.org/faq/.

See Also

multi_impute_trees

Examples

```
path <- tempfile(fileext = ".tre")
ape::write.tree(ape::rtree(10L), path)
tree <- read_tree(path)
```

save_pigauto *Save a fitted pigauto model*

Description

Serialises a pigauto_fit object to disk, including the torch model state. The saved file can be loaded on any machine with the same pigauto version.

Usage

```
save_pigauto(fit, path, compress = TRUE)
```

Arguments

fit	pigauto_fit object, or a pigauto_result from impute (its $fit is saved).
path	character. File path to save to (recommended extension: .pigauto).
compress	logical. Use gzip compression (default TRUE).

Details

Standard saveRDS() does not work for pigauto fits because torch tensor states cannot be serialised by R's native mechanism. This function converts the model state to a portable raw format before saving.

Value

Invisible path.

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
traits$Mass[seq_len(3L)] <- NA_real_
fit <- impute(traits, tree, epochs = 5L, verbose = FALSE)$fit
path <- tempfile(fileext = ".pigauto")
save_pigauto(fit, path)
fit2 <- load_pigauto(path)
```

`simulate_benchmark` *Run a simulation benchmark for pigauto*

Description

Generates trait data under various evolutionary models, introduces missing data, fits both the Brownian motion baseline and the full pigauto GNN, and compares performance. This is the recommended way to assess pigauto on data with known properties before applying it to real data.

Usage

```
simulate_benchmark(
  n_species = 100L,
  n_traits = 4L,
  scenarios = c("BM", "OU", "regime_shift", "nonlinear", "mixed"),
  missing_frac = 0.25,
  n_reps = 3L,
  epochs = 500L,
  verbose = TRUE,
  seed = NULL,
  gnn = TRUE,
  ...
)
```

Arguments

`n_species`	integer. Number of tips in the simulated tree (default 100).
`n_traits`	integer. Number of continuous traits (default 4). Ignored for `scenario = "mixed"`, which generates a fixed trait set.
`scenarios`	character vector. Subset of `c("BM", "OU", "regime_shift", "nonlinear", "mixed")`. Default runs all.
`missing_frac`	numeric. Fraction of observed cells held out (default 0.25).
`n_reps`	integer. Number of replicate trees per scenario (default 3).
`epochs`	integer. Maximum GNN training epochs (default 500).
`verbose`	logical. Print progress (default `TRUE`).
`seed`	optional integer. When supplied, makes the generated benchmark replicates reproducible.
`gnn`	logical. Whether the pigauto arm trains the GNN. Default `TRUE`, so this function keeps comparing the baseline with the GNN-corrected model even though `fit_pigauto` and `impute` default to `gnn = FALSE`.
`...`	additional arguments passed to `fit_pigauto`.

Details

Available scenarios:

`"BM"` Pure Brownian motion – the baseline is exact, so the GNN should tie or slightly improve via inter-trait correlations.

"OU" Ornstein-Uhlenbeck – stabilising selection constrains variation. BM over-estimates evolutionary variance.

"regime_shift" Two-regime BM – clade-specific optima create bimodal distributions that BM cannot capture.

"nonlinear" Non-linear inter-trait relationships – the GNN's multi-layer message passing can capture quadratic and interaction effects that BM's linear covariance misses.

"mixed" Mixed trait types: 2 continuous + 1 binary + 1 categorical (3 levels). Tests the full type pipeline.

Value

An object of class "pigauto_benchmark" with:

results data.frame with columns: scenario, rep, method, trait, type, metric, value, n_test.

summary data.frame averaged across replicates.

scenarios character vector of scenarios run.

n_reps integer.

n_species integer.

Examples

```
bench <- simulate_benchmark(n_species = 20L, n_traits = 2L,
                            scenarios = "BM", epochs = 5L, n_reps = 1L,
                            verbose = FALSE, gnn = FALSE)
bench$summary
plot(bench)
```

---

simulate_non_bm *Simulate non-BM trait data for benchmarking*

Description

Generates species trait data under models that deviate from Brownian Motion. Supports OU (stabilising selection), regime shifts (clade-specific optima), and non-linear inter-trait correlations.

Usage

```
simulate_non_bm(
  tree,
  n_traits = 4,
  scenario = c("OU", "regime_shift", "nonlinear"),
  alpha = 2,
  theta = 0,
  sigma = 1,
  shift_magnitude = 2,
  seed = NULL
)
```

Arguments

tree	Object of class "phylo".
n_traits	Integer. Number of traits to simulate (default 4).
scenario	Character. One of "OU", "regime_shift", "nonlinear".
alpha	Numeric. OU pull strength (only for "OU").
theta	Numeric. OU optimum (only for "OU").
sigma	Numeric. BM diffusion rate.
shift_magnitude	Numeric. Regime shift size in SD units (only for "regime_shift").
seed	Integer seed or NULL.

Value

A data.frame with species as rownames.

Examples

```
tree <- ape::rtree(12)
traits <- simulate_non_bm(tree, n_traits = 2, scenario = "OU", seed = 1)
head(traits)
```

---

suggest_next_observation

Suggest model-based candidate observations

Description

For a fitted pigauto_result (returned by impute), compute model-based BM/label-propagation proxy reductions in predictive variance or entropy across currently-missing cells if each candidate cell were observed next. Use the returned ranking as sampling-design guidance under those model assumptions; it is not an optimal-design guarantee or a demonstrated field gain.

Usage

```
suggest_next_observation(
  result,
  top_n = 10L,
  by = c("cell", "species"),
 types = c("continuous", "count", "ordinal", "proportion", "binary", "categorical",
    "zi_count", "multi_proportion")
)
```

Arguments

result	A pigauto_result object returned by impute. Must have been produced from single-obs data; multi-obs inputs error with a clear message.
top_n	integer, default 10L. Number of suggestions to return (descending by delta).

by — character, one of "cell" (default) or "species". "cell" returns individual (species, trait) pairs. "species" aggregates by species (summing reductions across the species' currently-missing traits).

types — character vector of pigauto trait types to include. Default includes all eight supported types: continuous, count, ordinal, proportion, binary, categorical, zi_count (added v2, 2026-05-01), multi_proportion (added v2).

Details

Two metrics are supported, dispatched by trait type:

Continuous-family traits (continuous, count, ordinal, proportion) use the BM variance-reduction formula. The variance- reduction formula is derived from a Sherman-Morrison rank-1 inverse update on the BM conditional MVN: adding species s to the observed set updates the inverse correlation matrix by a known closed form. For each candidate cell (s, t),

$$\Delta V(s,t) = \sigma_t^2 \sum_{i \in \mathrm{miss}_t} \frac{D_{ik}^2}{\alpha_k}$$

where $D = R_{mm} - R_{mo}R_{oo}^{-1}R_{om}$ is the residual matrix at currently-missing cells, $\alpha_k = D_{kk}$ is the current relative leverage of cell k, and σ_t^2 is the REML BM variance for trait t.

Discrete traits (binary, categorical) use a label- propagation expected-entropy-reduction formula. The current LP probability at species i is $p_i = \mathrm{sim}[i, \mathrm{obs}]y_{\mathrm{obs}} / \sum \mathrm{sim}[i, \mathrm{obs}]$, with entropy $H(p_i) = -\sum_k p_{i,k} \log p_{i,k}$. After observing s_{new} with unknown class y_{new}, the new LP probability has a closed form, and the expected entropy is averaged over $P(y_{\mathrm{new}})$ = current LP estimate at s_{new}. Total expected entropy reduction sums across all currently-missing cells (the entropy at s_{new} itself drops to 0).

Variance and entropy are NOT directly comparable. The output sorts within each metric and the cross-metric ordering by delta is approximate. When you want a strict ranking, filter by metric first.

Reductions are summed across the included traits for each species when by = "species", supporting the typical use case where measuring a species observes all of its currently-missing traits at once. At by = "species", the per-trait variance and entropy reductions are summed separately into delta_var_total and delta_entropy_total columns; the delta column is whichever is non-NA (or delta_var_total when both are populated). Cross-type species-level ranking is approximate – see the variance-vs- entropy caveat above.

The ranking is conditional on the fitted BM or label-propagation proxy and its missing-data assumptions. It does not establish that collecting a selected observation improves a future field study or downstream analysis.

zi_count (v2): observing a missing zi_count cell reveals the gate value (entropy reduction at the gate column, computed via the LP binary formula) AND, with probability p_{gate} (current LP estimate at s_{new}), reveals a magnitude (variance reduction at the magnitude column, computed via the BM Sherman-Morrison formula on the gate=1 subset). Output rows for zi_count populate BOTH delta_var_total (= expected magnitude variance reduction = $p_{\mathrm{gate}} \times \Delta V_{\mathrm{mag}}$) AND delta_entropy_total (= gate entropy reduction). metric is set to "variance" so the row sorts on the magnitude scale; delta_entropy_total is available for users who care about gate-uncertainty separately.

multi_proportion (v2): observing a row reveals all K simplex components simultaneously. Per-component variance reductions are computed via BM Sherman-Morrison on each CLR-z latent column, summed across components. metric is "variance"; delta_var_total is the K-component sum.

Value

A data.frame of class "pigauto_active". Columns when by = "cell": species, trait, type, metric ("variance" or "entropy"), delta, delta_var_total (NA for discrete rows), and delta_entropy_total (NA for continuous rows), sorted by delta descending. When by = "species": species, delta_var_total, delta_entropy_total, n_traits_missing, sorted by the SUM of available metrics.

See Also

impute

Examples

```
data(avonet300, tree300, package = "pigauto")
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
traits$Mass[seq_len(3L)] <- NA_real_
res <- impute(traits, tree, epochs = 5L, verbose = FALSE)
suggest_next_observation(res, top_n = 5L)
suggest_next_observation(res, top_n = 5L, by = "species")
```

---

summary.pigauto_fit *Summary method for pigauto_fit objects*

Description

Prints a formatted evaluation table including per-trait metrics, gate calibration, and conformal coverage. Requires the original pigauto_data object to compute test-set performance.

Usage

```
## S3 method for class 'pigauto_fit'
summary(object, ..., data = NULL)
```

Arguments

object	pigauto_fit object.
...	ignored.
data	pigauto_data object used for fitting (optional; when NULL the summary skips per-trait test metrics).

Value

Invisibly returns the evaluation data.frame (or NULL if data is not supplied).

Examples

```
data(avonet300, tree300)
tree <- ape::keep.tip(tree300, tree300$tip.label[seq_len(30L)])
traits <- avonet300[match(tree$tip.label, avonet300$Species_Key),
                    c("Mass", "Wing.Length"), drop = FALSE]
rownames(traits) <- tree$tip.label
pd <- preprocess_traits(traits, tree)
splits <- make_missing_splits(pd$X_scaled, trait_map = pd$trait_map)
fit <- fit_pigauto(pd, tree, splits = splits, epochs = 5L,
                   verbose = FALSE)
summary(fit, data = pd)
```

---

summary.pigauto_result

Summarise a pigauto result

Description

Summarise a pigauto result

Usage

```
## S3 method for class 'pigauto_result'
summary(object, ...)
```

Arguments

object	A pigauto_result.
...	Unused.

Value

A small summary_pigauto_result object containing input-check, completion, output-role, and evaluation summaries.

---

tree_full

Example bird phylogeny for the species in avonet_full

Description

An object of class 'phylo' from the **ape** package. It is posterior sample member 69 from the Hackett-backbone half of BirdTree's 100-tree bird collection, pruned to the species present in avonet_full rather than the 300-species example subset. The MIT software licence for **megatrees** does not establish rights for the underlying BirdTree data; see the installed 'NOTICE'. This object is included for examples, not as an MCC or consensus tree or a recommended tree for substantive analysis.

Usage

```
tree_full
```

Format

An object of class phylo with 9,993 tips.

Details

Row order in avonet_full matches tip order in tree_full: all(avonet_full$Species_Key == tree_full$tip.label) returns TRUE.

Source

Li (2026), **megatrees** 1.0.0; member 69 of megatrees::get_tree_bird_n100() is from the Hackett-backbone set. The BirdTree distribution is described by Jetz et al. (2012).

References

Hackett SJ, Kimball RT, Reddy S, et al. (2008). A phylogenomic study of birds reveals their evolutionary history. *Science*, 320(5884), 1763-1768. doi:10.1126/science.1157704.

Jetz W, Thomas GH, Joy JB, Hartmann K, Mooers AO (2012). The global diversity of birds in space and time. *Nature*, 491, 444-448. doi:10.1038/nature11631.

See Also

tree300, avonet_full

---

tree300 *Example bird phylogeny for the 300 species in* avonet300

Description

An object of class 'phylo' from the **ape** package. It is posterior sample member 69 from the Hackett-backbone half of BirdTree's 100-tree bird collection, retrieved through the **megatrees** package and pruned to the 300 species in avonet300. The MIT software licence for **megatrees** does not establish rights for the underlying BirdTree data; see the installed 'NOTICE'. This object is included for examples, not as an MCC or consensus tree or a recommended tree for substantive analysis.

Usage

```
tree300
```

Format

An object of class phylo with 300 tips.

Source

Li (2026), **megatrees** 1.0.0; member 69 of megatrees::get_tree_bird_n100() is from the Hackett-backbone set. The BirdTree distribution is described by Jetz et al. (2012).

References

Hackett SJ, Kimball RT, Reddy S, et al. (2008). A phylogenomic study of birds reveals their evolutionary history. *Science*, 320(5884), 1763-1768. doi:10.1126/science.1157704.

Jetz W, Thomas GH, Joy JB, Hartmann K, Mooers AO (2012). The global diversity of birds in space and time. *Nature*, 491, 444-448. doi:10.1038/nature11631.

`trees300` *50 posterior phylogenies for the 300 species in* `avonet300`

Description

A `multiPhylo` list of 50 phylogenetic trees sampled from the 100-tree BirdTree bird collection retrieved through **megatrees**. The collection's first 50 trees use the Ericson backbone and its next 50 use the Hackett backbone. The fixed sample generated with seed 42 contains 28 Ericson and 22 Hackett trees, each pruned to the 300 species in `avonet300`. Topologies and branch lengths vary across this posterior sample. The MIT software licence for **megatrees** does not establish rights for the underlying BirdTree data; see the installed '`NOTICE`'.

Usage

```
trees300
```

Format

An object of class `multiPhylo` containing 50 `phylo` objects, each with 300 tips.

Details

Use with `multi_impute_trees` for experimental sensitivity of point imputations to posterior-tree choice. Tree uncertainty is not supported by the analysis-aware MI backend, and these stochastic datasets are not validated for downstream inference.

Source

Li (2026), **megatrees** 1.0.0; trees pruned from `megatrees::get_tree_bird_n100()`. The source collection and its Ericson and Hackett backbones are described by Ericson et al. (2006), Hackett et al. (2008), and Jetz et al. (2012).

References

Ericson PGP, Anderson CL, Britton T, et al. (2006). Diversification of Neoaves: integration of molecular sequence data and fossils. *Biology Letters*, 2(4), 543-547. doi:10.1098/rsbl.2006.0523.

Hackett SJ, Kimball RT, Reddy S, et al. (2008). A phylogenomic study of birds reveals their evolutionary history. *Science*, 320(5884), 1763-1768. doi:10.1126/science.1157704.

Jetz W, Thomas GH, Joy JB, Hartmann K, Mooers AO (2012). The global diversity of birds in space and time. *Nature*, 491, 444-448. doi:10.1038/nature11631.

See Also

`tree300`, `avonet300`, `multi_impute_trees`

`with_imputations` *Fit a downstream model on every imputed dataset*

Description

Apply a user-supplied model-fitting function `.f` to each of the `M` complete datasets stored in an analysis-aware MI object and return the list of fits. This is the supported fitting entry for `multi_impute_analysis()` and for `multi_impute()` with `draws_method = "posterior"`. Prediction-diagnostic (`"conformal"`, `"mc_dropout"`) and posterior-tree sensitivity completions stop with an actionable error before any model is fitted. The workflow is experimental and limited to fixed-effect coefficients and their covariance matrices.

Usage

```
with_imputations(
  mi,
  .f,
  ...,
  .progress = interactive(),
  .on_error = c("continue", "stop")
)
```

Arguments

`mi`	A `pigauto_analysis_mi` object returned by `multi_impute_analysis()`, or a `pigauto_posterior_mi` object returned by `multi_impute()` with `draws_method = "posterior"` (and `param_uncertainty = "full"`, the default, or `"both"`). Legacy `pigauto_mi` objects, diagnostic completions, tree-sensitivity completions, plug-in posterior draws, and bare dataset lists are not supported fitting inputs.
`.f`	A function of the form `function(dataset, ...)` that fits a model to one complete data.frame and returns a model object. `pool_mi()` supplies automatic fixed-effect adapters for its documented model classes. Other classes require `coef()` and `vcov()` methods that return compatible fixed-effect quantities, or explicit extractor functions supplied to `pool_mi()`; extractability does not imply that an unlisted analysis model has passed the analysis-aware validation gate.
`...`	Additional arguments passed to `.f` for every imputation.
`.progress`	Logical. Show a text progress indicator (default `TRUE` in interactive sessions).
`.on_error`	One of `"continue"` (default) or `"stop"`. When `"continue"`, errors from `.f` are captured per imputation and the loop proceeds; a warning at the end summarises failures. When `"stop"`, the first error aborts the entire run.

Value

A list of length `M` with class `"pigauto_mi_fits"`. Each element is either a model fit or, if `.f` errored on that imputation and `.on_error = "continue"`, an object of class `"pigauto_mi_error"` containing the captured condition. `pool_mi()` filters error elements automatically.

Posterior draws and congeniality

The `draws_method = "posterior"` completions come from a linear-Gaussian phylogenetic mixed model of the imputed traits only, on the scale where they were imputed (the log scale for traits log-transformed by `log_transform`). They are proper imputations for analyses that are linear in the imputed traits on that scale and whose variables are all among the imputed traits, for example a phylogenetic regression of one imputed trait on others (on the log scale for log-transformed traits). Not covered: covariates from outside the imputed traits, nonlinear terms (such as squares) or interactions among imputed traits, and analysing a log-transformed trait on its raw scale. The formula passed to `.f` is not checked; this is the analyst's responsibility. Plug-in draws from `posterior_control$param_uncertainty = "none"` (validation only) are refused.

See Also

`multi_impute_analysis()`, `pool_mi()`

Examples

```
dat <- data.frame(y = stats::rnorm(30L), x = stats::rnorm(30L),
                  z = stats::rnorm(30L))
dat$x[seq(3L, 30L, by = 5L)] <- NA_real_
mi <- multi_impute_analysis(
  data = dat, formula = y ~ x + z, missing = "x",
  model = "lm", m = 2L
)
fits <- with_imputations(mi, function(d) stats::lm(y ~ x + z, data = d))
pool_mi(fits)
```

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